

F/A-67 Condor Critical Design Review



Group 01- No Thai Grubbin
Michael Chen, Charles Choi, Cristina Erskine, James Gold,
Santiago Ramos-Assam, Thomas Sheridan
December 5, 2025

Request for Proposal

Next Generation Carrier-Based Strike Fighter

- Replacement for F/A-18E/F aircraft
- 500 units with 2035 initial operating capability
- 25-year service life

General Design Requirements

- Use existing production engines and currently available materials
- Technology readiness level (TRL) 6 subsystems
- Internal carriage of ordnance desired

Objective

- Maximize combat performance with comparable cost to F/A-18E/F
- Flyaway unit cost, 500 unit production run: < 113 \$M 2025

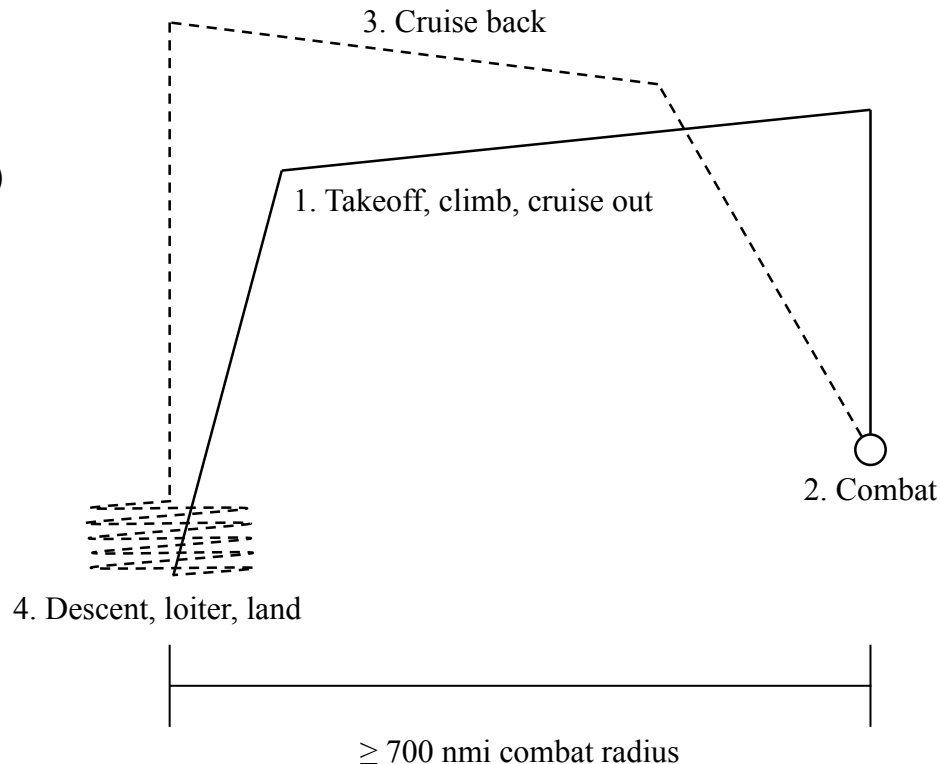
Air to Air Combat Mission

Minimum Performance Requirements

- 700 nmi combat radius
- 2 min combat at maximum thrust (10,000 ft)
- 8 deg/s sustained turn rate (20,000 ft)
- Mach 1.6 dash speed (30,000 ft)
- 7g vertical load factor (mid mission weight)

Payload

- 6x AIM-120C
- 2x AIM-9X



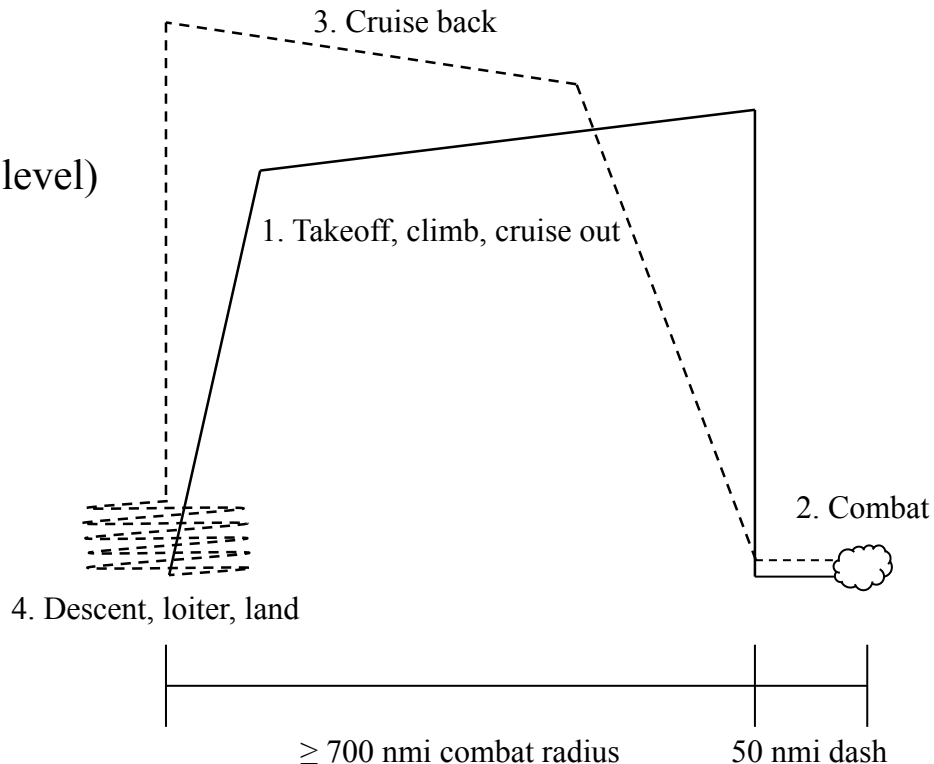
Strike Mission

Minimum Performance Requirements

- 700 nmi combat radius
- 2x 50 nmi dashes at intermediate thrust (sea level)
- Mach 0.85 dash speed (sea level)
- 7g vertical load factor (mid mission weight)

Payload

- 4x MK-83 JDAM
- 2x AIM-9X



Design Summary

Dimensions

- Length: 49.8 ft
- Span: 48.75 ft, Stowed span: 35 ft
- Height: 17.87 ft
- Planform area: 648 ft²
- Spot factor: 1.11

Weights

- Empty weight: 31,207 lb
 - Avionics: 2500 lb
- Payload: 2,460 - 4,424 lb
- Fuel: 26,959 lb
- Takeoff weight: 60,826 lb

Configuration

Crew: 1

Wing:

- Swept
- Supercritical airfoil
- Leading edge flaps
- Slotted flaps and flaperons

Empennage:

- Swept, biconvex airfoil
- All moving horizontal stabilator
- Twin vertical stabilizers

Power Plant:

- 2x F110-GE-132

Post PDR Design Refinement

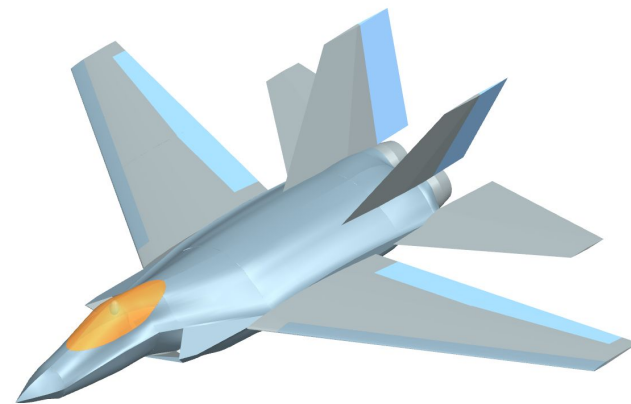
Improved Models

- Mission Profiles
 - Optimal cruise-climb
 - Fuel burn estimation
- Empty Weight
 - Component weight estimation
- Aerodynamics
 - Parasitic drag estimation
 - $C_{L,max}$ prediction

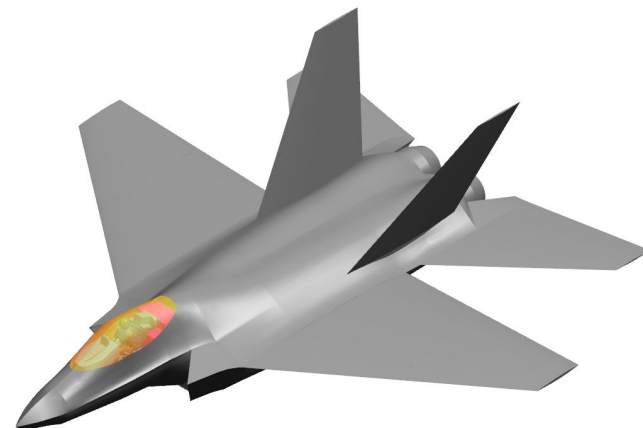
Design Trades

- Internal vs. external ordnance storage
- Mission performance vs. planform sizing

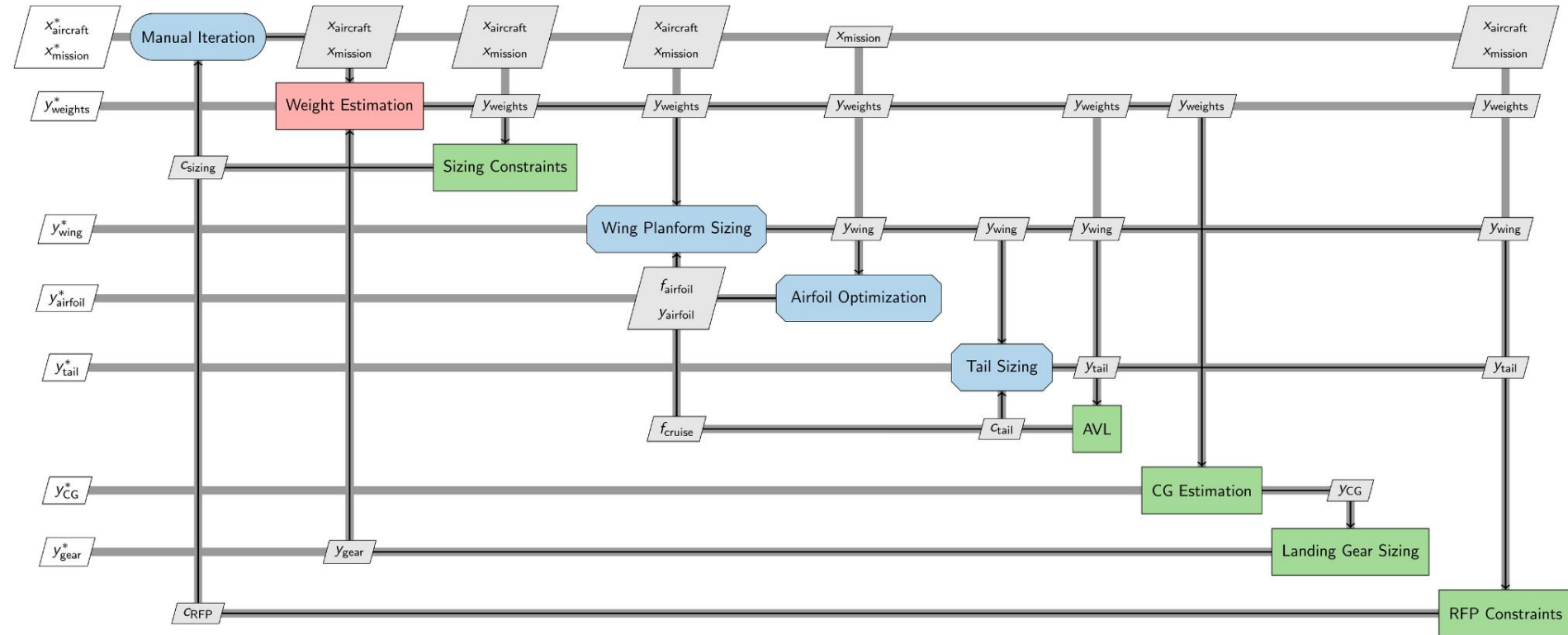
PDR F/A-67



CDR F/A-67



Computational Procedure



Configuration Selection

Preliminary Sizing Trends

- Air-to-air mission has greater takeoff weight vs. strike mission
- Higher mission performance increases takeoff weight

RFP Design Drivers

- Catapult launch requirements limit takeoff weight
- Aircraft dimensions requirements limit span and planform area
- Supersonic dash constraint requires high afterburning T/W



Design for Objective

- Slotted flaps increase $C_{L,max}$, increasing catapult launch takeoff weight
- Wing area minimized to reduce aircraft size and weight
- 2 engines selected to provide sufficient T/W

Ordnance Storage Trade Study

External Configurations

- Air-to-air
 - 2 pylons, 2 racks
 - 4x AIM-120C on racks
 - 2x AIM-120C fuselage mounted
 - 2x AIM-9X wingtip mounted
- Strike
 - 2 pylons, 2 racks
 - 4x MK-83 JDAM on racks
 - 2x AIM-9X wingtip mounted

Conclusions

- Significant fuselage drag reduction required
- Externals penalize supersonic wave drag

Results (ΔC_{D0} in drag counts)

	0 External Fuel Tanks	2 External Fuel Tanks
0 External Weapons	0.00	17.39
2 AIM-9X's	1.31	18.70
6 AIM-120Cs	21.94	39.33
4 MK-83 JDAM	22.88	40.28
Air-to-air	23.25	40.64
Strike	24.19	41.59
Fuselage (internal)	74.50	—

Mission Performance Trade Study

Design Variables

- Combat radius
- Combat time
- Dash Mach number
- Wingspan
- Wing aspect ratio

Design Point Selection

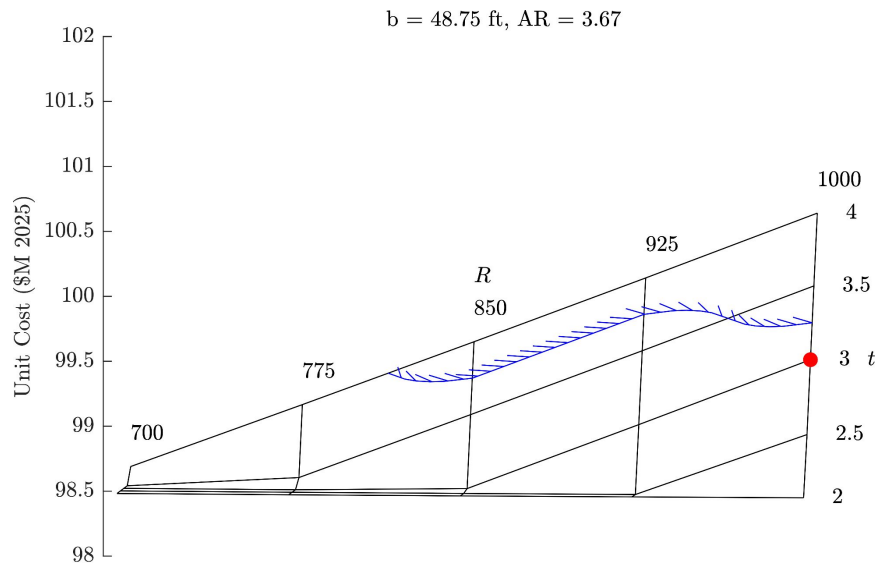
- Planform area $< 670 \text{ ft}^2$ (PDR design)
- Select wingspan, aspect ratio combination with highest mission performance
- Unit cost $< 113 \text{ \$M 2025}$

Results

- Span: 48.75 ft, Area: 648 ft^2 , $AR = 3.67$

Mach 1.65

Feasible designs below blue line



Mission Performance Trade Study

Refinement

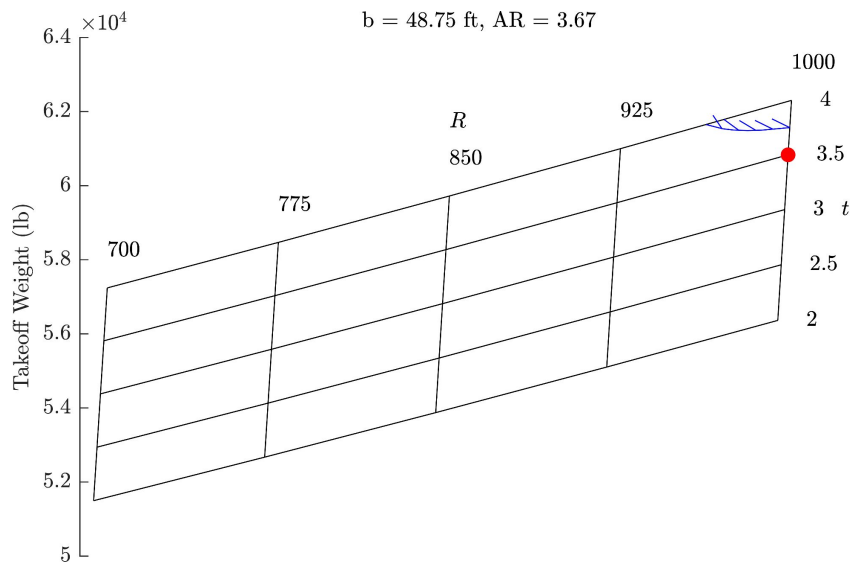
- Updated aircraft geometry and aerodynamics yields higher combat time
- Less weight increase from increasing combat radius

Air-to-air Mission Performance

- 1000 nmi desired combat radius
- 3.5 min combat time
- Mach 1.65 supersonic dash

Mach 1.65

Feasible designs below blue line



Thrust-to-Weight vs. Wing Loading Sizing Plot (Air-to-Air)

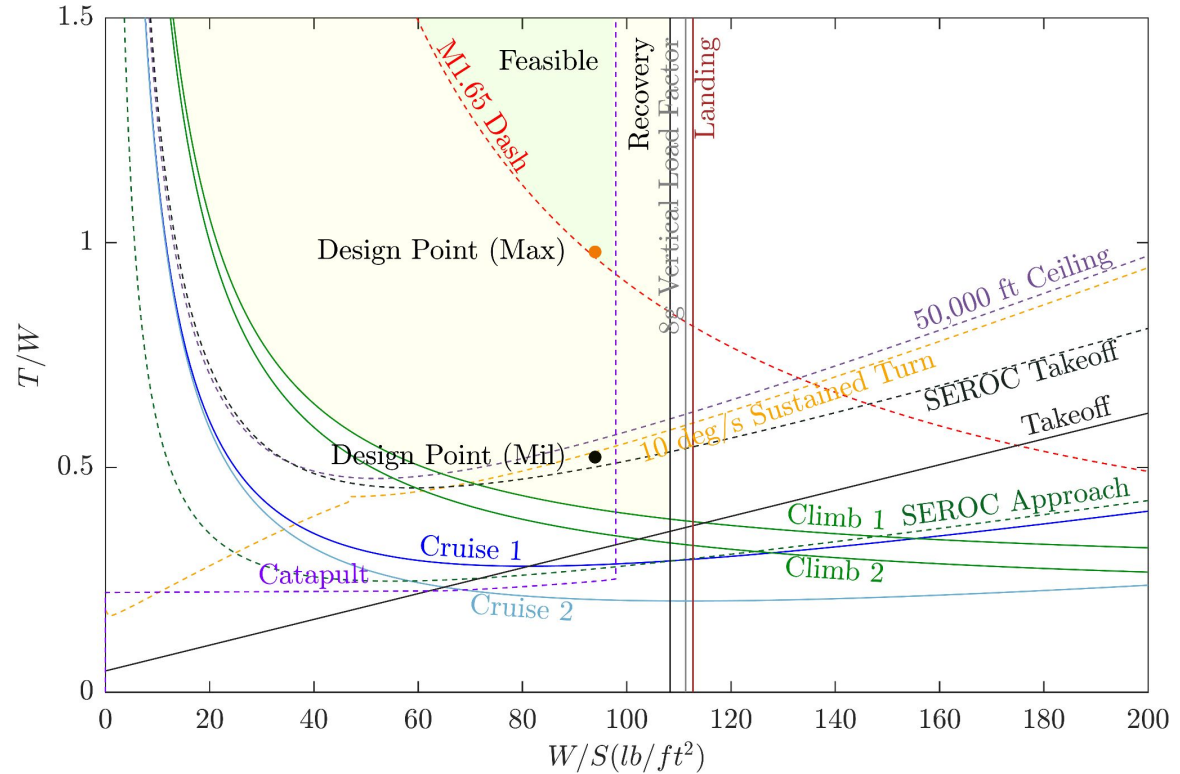
$$W/S = 94 \text{ lb/ft}^2$$

$$T/W (\text{max}) = 0.98$$

$$T/W (\text{mil}) = 0.52$$

Constrained by dash and
catapult launch

Design point meets
afterburning and dry
T/W requirement



Thrust-to-Weight vs. Wing Loading Sizing Plot (Strike)

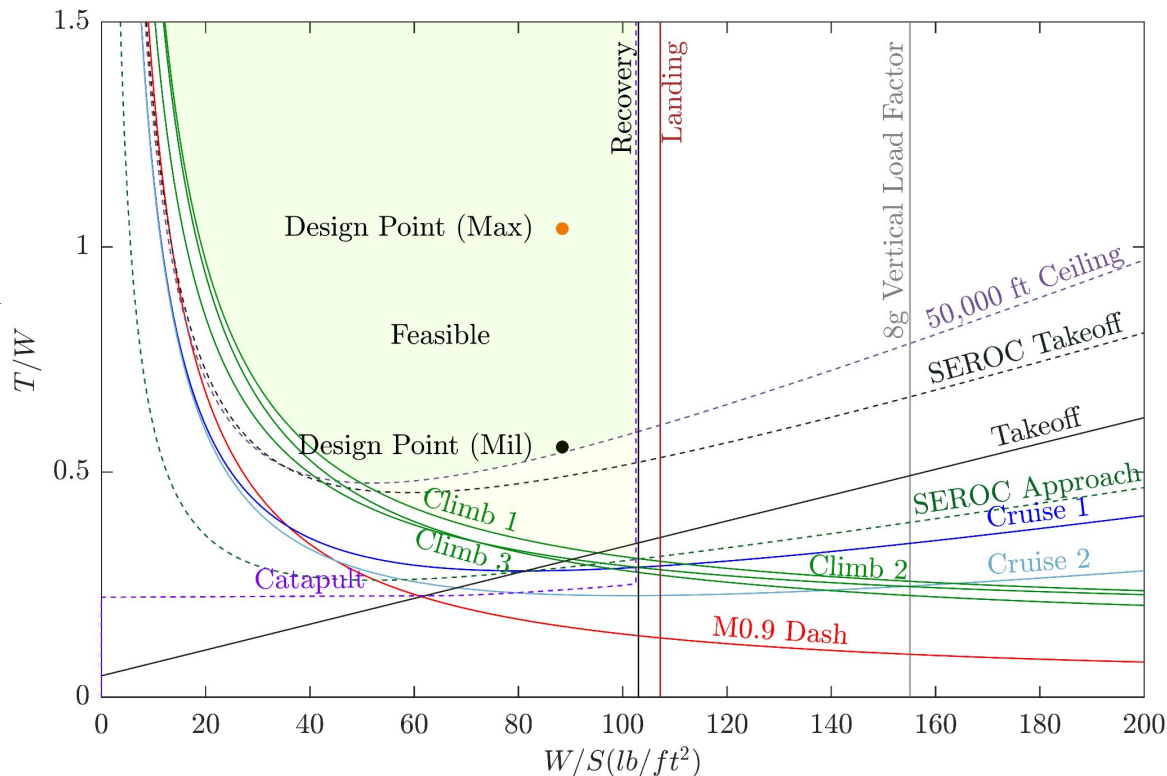
$$W/S = 88 \text{ lb/ft}^2$$

$$T/W (\text{max}) = 1.04$$

$$T/W (\text{mil}) = 0.55$$

Constrained by climb, conventional takeoff, and catapult launch

Design point meets
afterburning and dry
T/W requirement

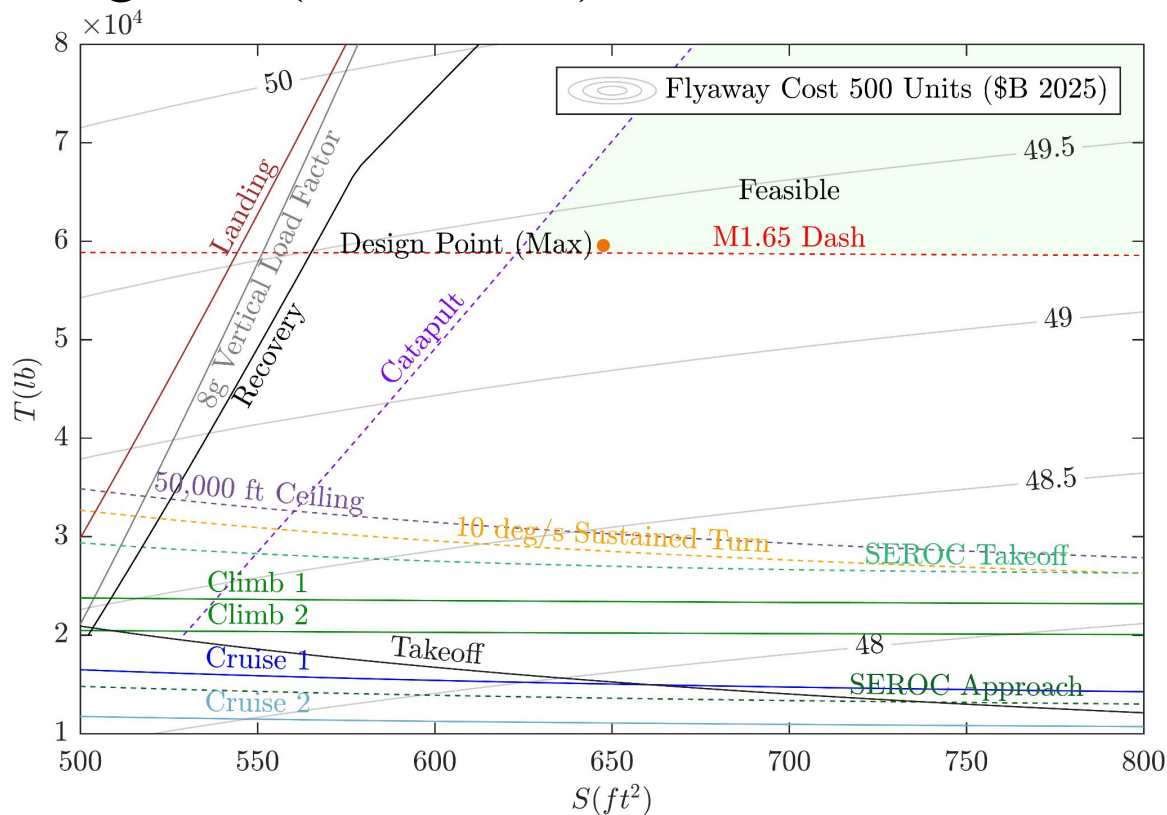


Thrust vs Wing Area Sizing Plot (Air-to-Air)

$$S = 648 \text{ ft}^2$$

$$T = 59,567 \text{ lb}$$

Design point meets
afterburning thrust
and cost requirement

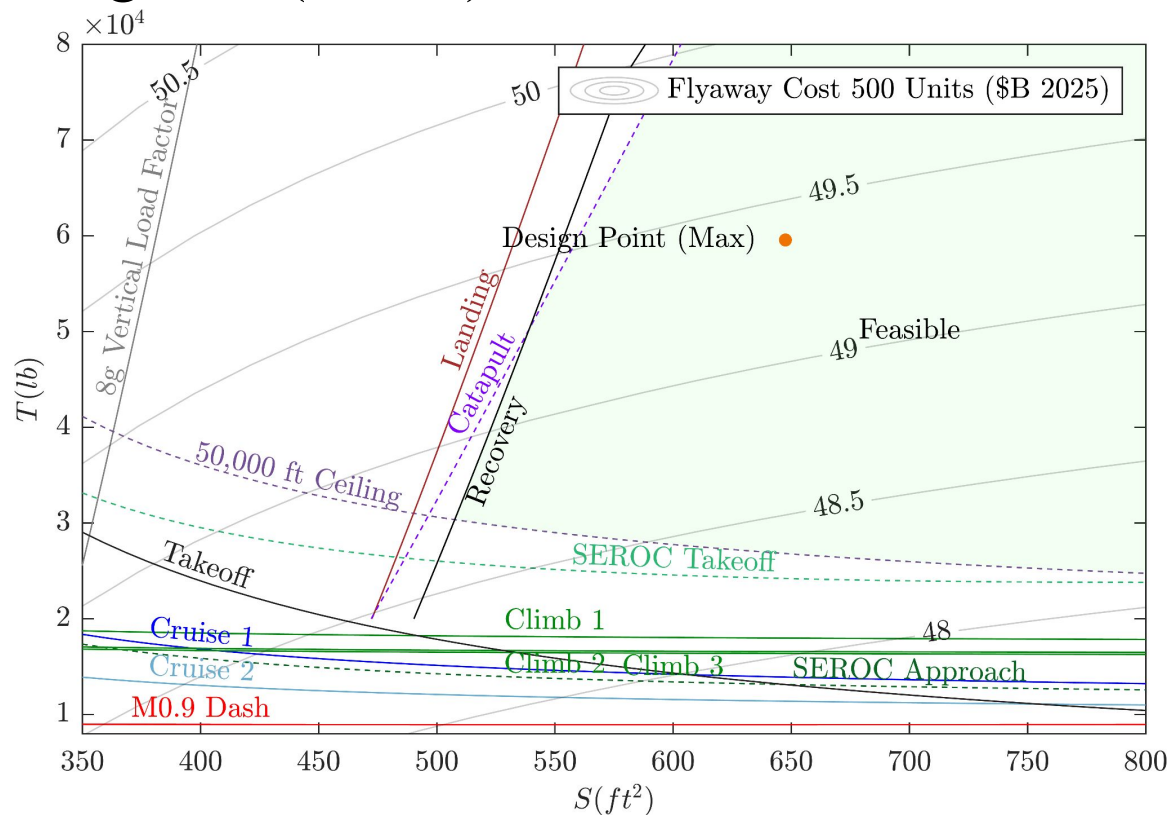


Thrust vs Wing Area Sizing Plot (Strike)

$$S = 648 \text{ ft}^2$$

$$T = 59,567 \text{ lb}$$

Design point meets
afterburning thrust
and cost requirement



Mission Performance and Weights

Air to air

- 1000 nmi combat radius
- 3.5 min combat time
- 12 deg/s sustained turn rate
- Mach 1.65 dash speed
- 8g vertical load factor

	Air to air	Strike
Empty weight (lb)	31,207	31,207
Payload weight (lb)	2,460	4,424
Fuel weight (lb)	26,959	21,425
Takeoff weight (lb)	60,826	57,257

Strike

- 1000 nmi combat radius
- Mach 0.9 dash speed
- 8g vertical load factor

Underlined parameters meet desired RFP values
All parameters exceed minimum requirements

Cost Breakdown (2025 USD)

Unit Flyaway Cost: 98.8 \$M

DOC/pay-nmi: 0.0138 \$

Total Engine Price: 18.8 \$M

Notes

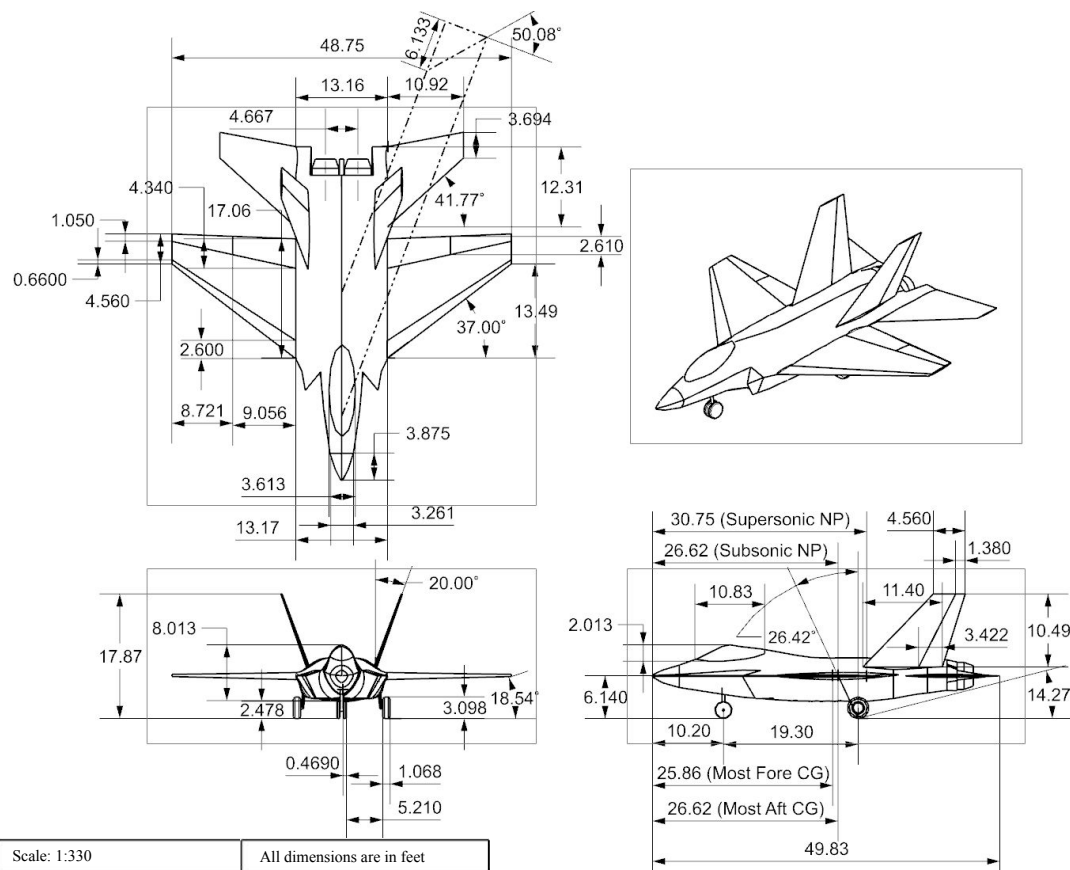
- Learning curve from Roskam [5] used of 73%
- Profit of 10% assumed for flyaway cost from Roskam
- F/A-18E/F Unit Flyaway Cost for 552 aircraft: 113 \$M

Component	Cost Per Aircraft Per Year (\$)
Fuel	1,030,000
Oil	5,140
Aircraft Maintenance	2,050,000
Engine Maintenance	729,000
Crew	1,110,000
Surface Treatments	203,000
Miscellaneous	1,620,000
Total DOC	6,740,000

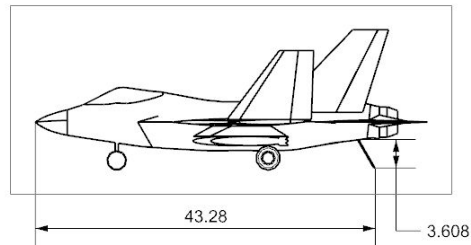
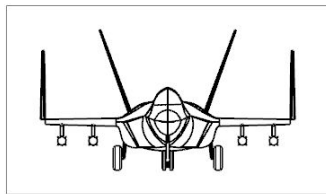
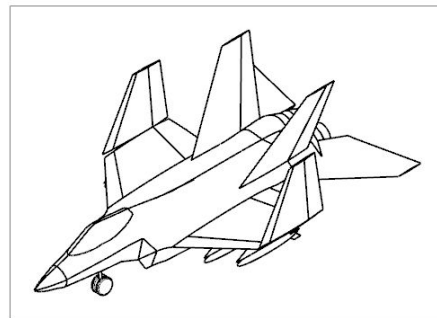
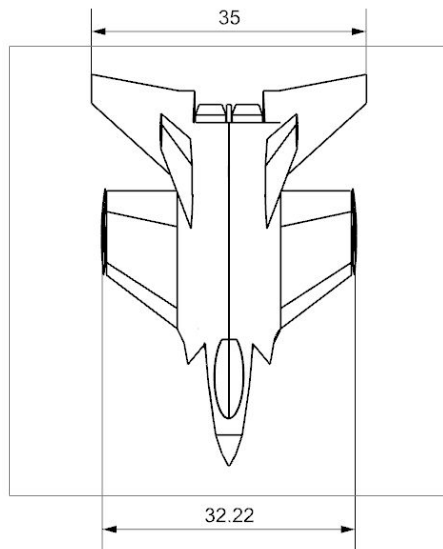
Comparison With Existing Aircraft

Aircraft	Empty Weight (lb)	Takeoff Weight (lb)	Length (ft)	Span (ft)	Reference Area (ft ²)	Aspect Ratio	T/W	W/S (lb/ft ²)	Combat Radius (nmi)	Flyaway Cost (\$M2025)
F/A-67	31,207	60,826	49.8	48.8	648	3.67	0.98	94	1,000	98.8
F/A-18E	32,081	66,000	60.2	44.7	500	3.5	0.65	132	462	113
F-35C	34,800	70,000	51.5	43.0	668	2.7	0.61	105	670	127
F-15E	37,500	81,000	63.8	42.8	608	3.0	0.73	133	687	61.4

3 View Drawings



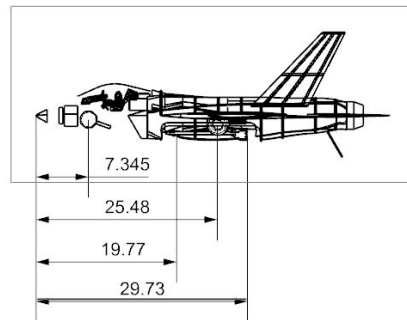
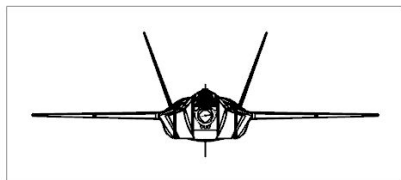
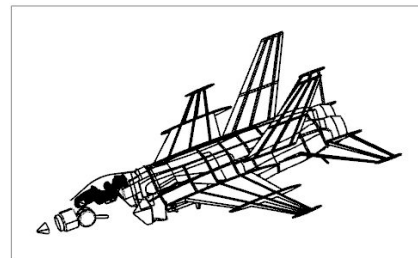
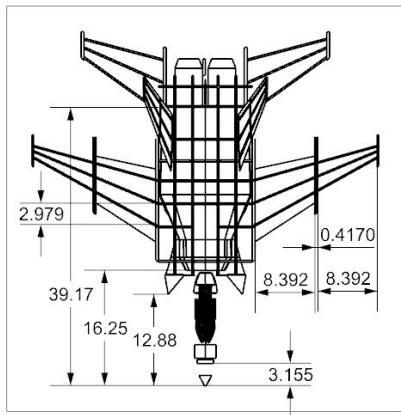
3 View Drawings (Stowed)



Scale: 1:295

All dimensions are in feet

Internal Layout



Scale: 1:330

All dimensions are in feet

Weapons Bays

Internal Storage

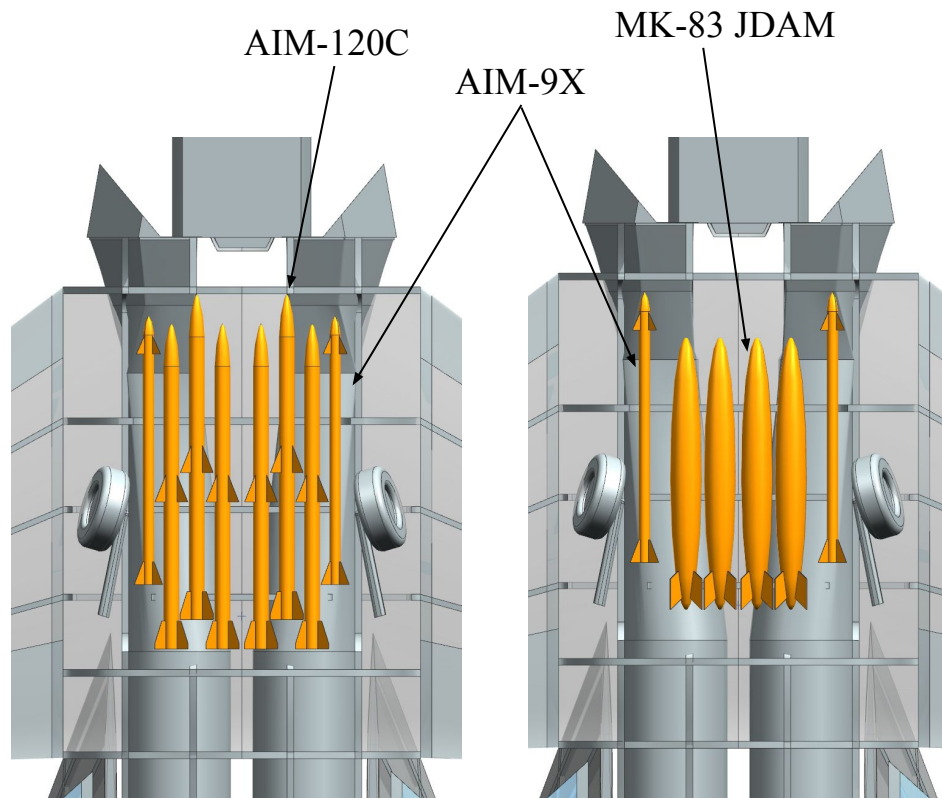
- Payload fully stored internally for reduced radar cross section and less drag

Air to Air Mission Payload

- 6x AIM-120C
- 2x AIM-9X

Strike Mission Payload

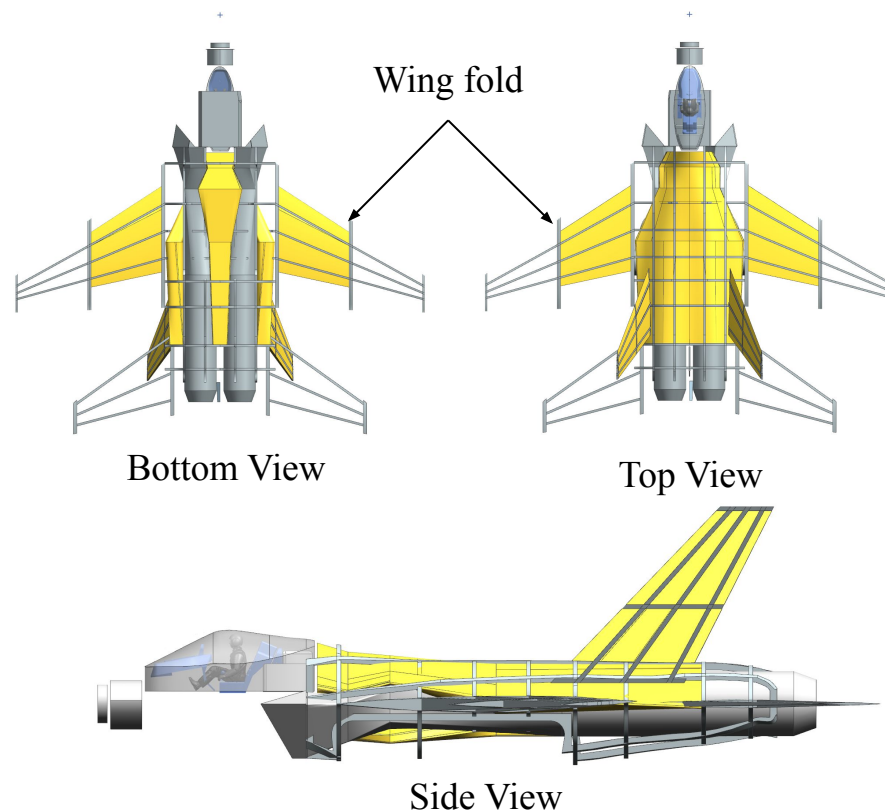
- 4x MK-83 JDAM
- 2x AIM-9X



Internal Weapons Bay

Fuel Tanks

Location	Fuel Volume (ft ³)	Fuel Weight (lb)
Wings	92.2	4,648.9
Vertical Tails	22.8	1,149.6
Fuselage	419.7	21,162.1
Total	534.7	26,960.6



Fuel Tanks shown in yellow

External Store Carriage

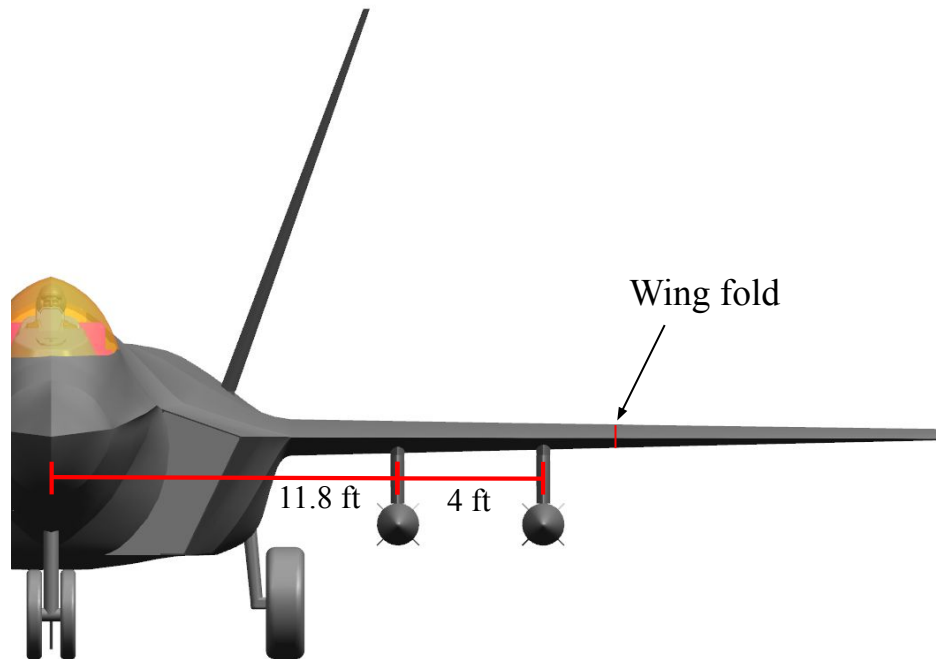
Requirements

- $\geq 10,000$ lb external storage
- $\geq 3,000$ lb per store

External Carriage

- 4 removable hardpoints

Store	Weight (lb)
480 gal drop tank	3,235
2x MK-83 JDAM	2,026
2x AIM-120C	696



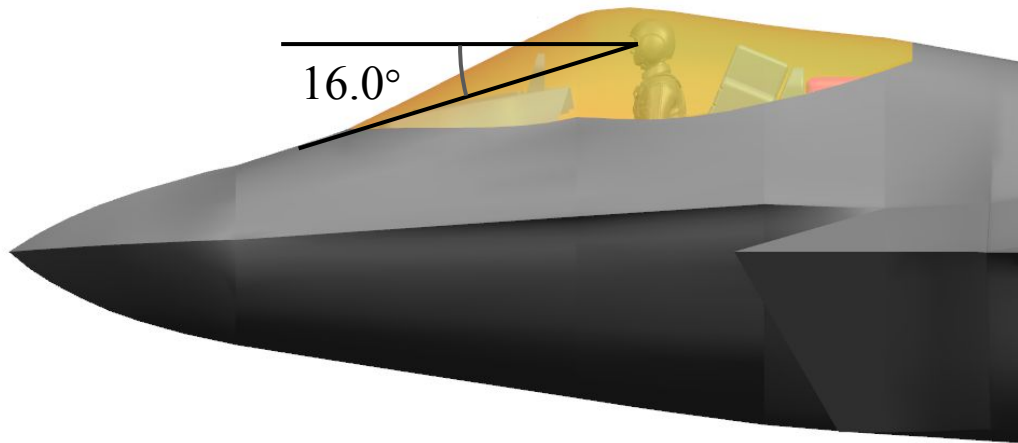
Pilot Sightlines

Crew

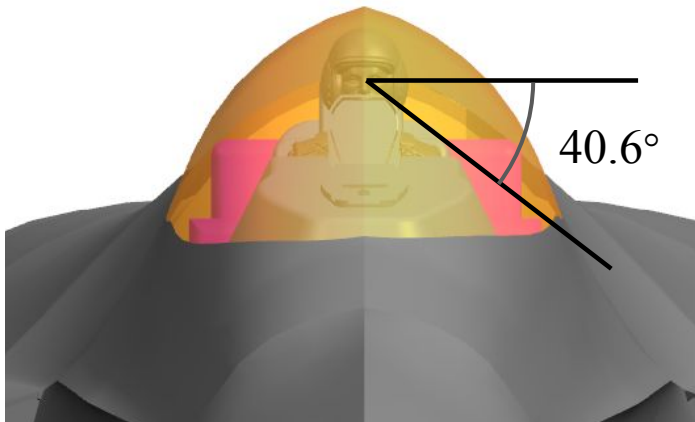
- 1 crew member
 - Reduced weight
 - Reduced volume

Pilot View Angles

- Overnose angle: 16.0°
- Over-the-side angle: 40.6°
- Grazing angle: 14.1°



Side View



Front View

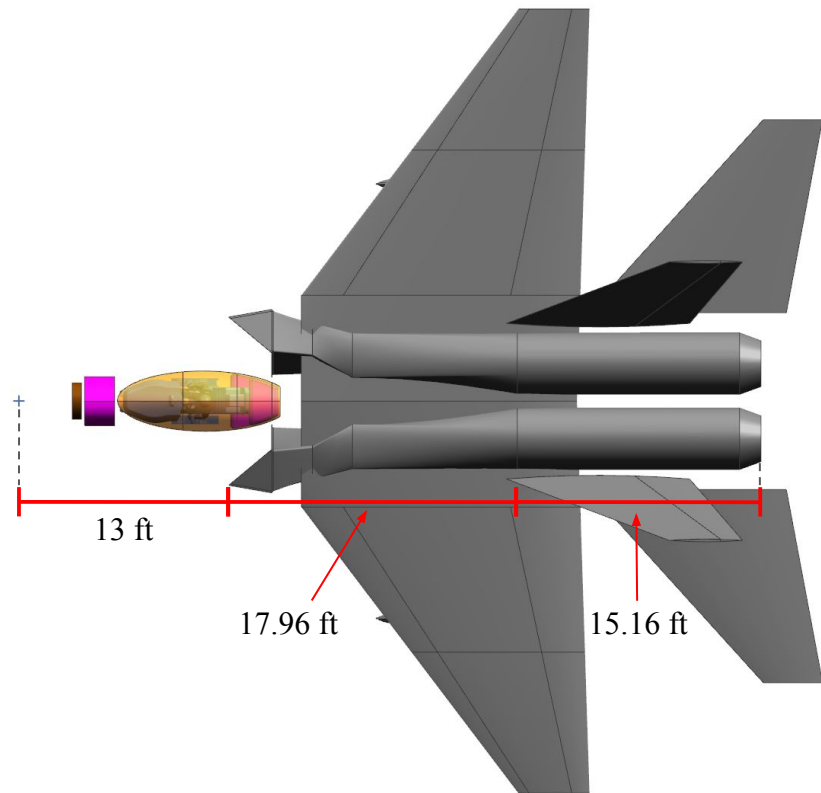
Power Plant

2x F110-GE-132 Engines

- Maximum Thrust: 33,093 SLS-lb per engine
- Military Thrust: 17,669 SLS-lb per engine
- Existing production design: used on F-16E/F
- Meets design thrust requirements with a lower dry TSFC than other comparable engines

Thrust Corrections

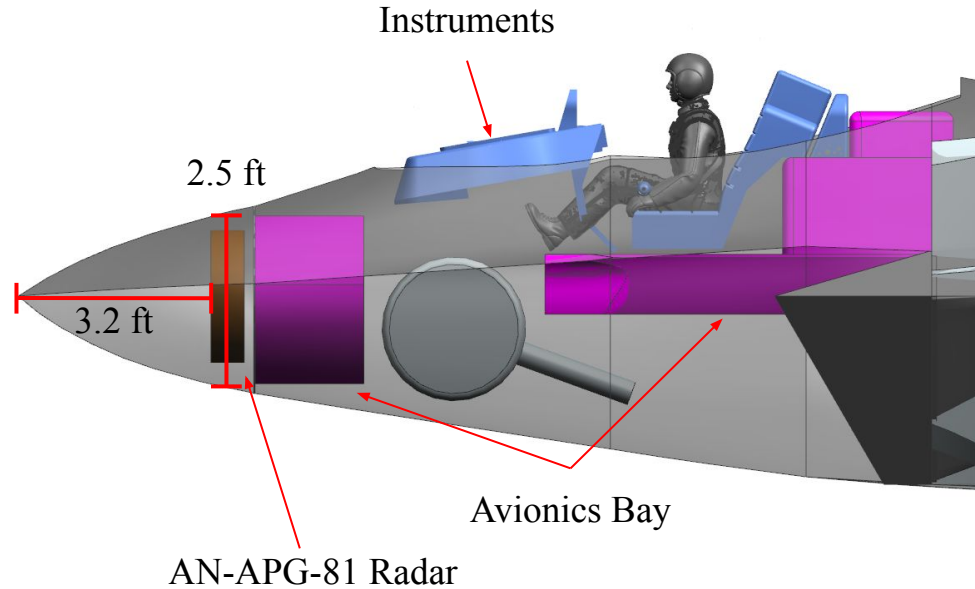
- Accounts for 10% installed thrust reduction
 - Inlet Pressure Loss: 1.35%
 - Engine Bleed: 8%
- Accounts for atmospheric thrust lapse and ram effects



Nose to inlets, inlets to engine, and engine length respectively.

Avionics and Subsystems

- Avionics Bays Volume: 61.4 ft³
- Avionics placed near the cockpit to reduce wire length
- AN/APG-81 Radar
 - Enemy tracking and ordnance guidance
- JPALS Approach & Landing System
 - Precision guidance for landing on carriers
- Quadruple redundant fly-by-wire system
 - Ensures aircraft is stable while being maneuverable
- Advanced Mission Computer
 - Modular system that processes and controls a lot of avionics



Additional GFE

- ADVIEW
 - Replaces the AN/ALQ-214 IDECM and AN/ALR-67
 - Counters enemy missiles with jammers and decoys
 - Detects radio emissions of enemy radar
- Genesis III
 - Helmet provides flight data, night vision, and 360° exterior view
- OBOGS
 - Oxygen system allows for high altitudes and high-G maneuvers
- Ejection Seat



Pilot wearing Genesis III Helmet [15]

Wing and Tail Trade Study

Procedure

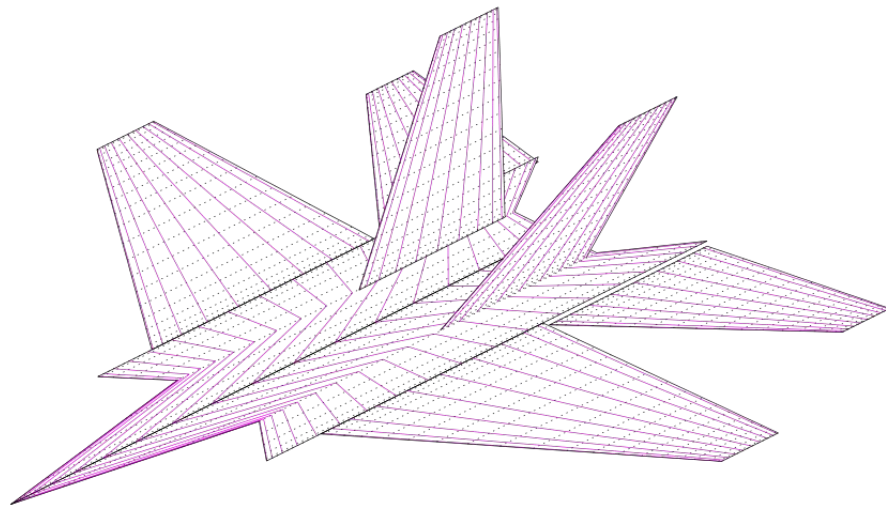
- Minimize total cruise drag using AVL
- Vary taper ratio, sweep angle, twist angle, wing x location, and stabilator moment arm
- Assume zero root incidence

Constraints

- Wing and stabilator must not overlap
- Zero static margin

Results

- Taper ratio of 0.2 minimizes parasitic drag
- 30° sweep optimal, higher sweep increases wave drag and reduces $C_{L,max}$
- 2.7° washout optimal for this wing



Wing Planform Design

Geometry

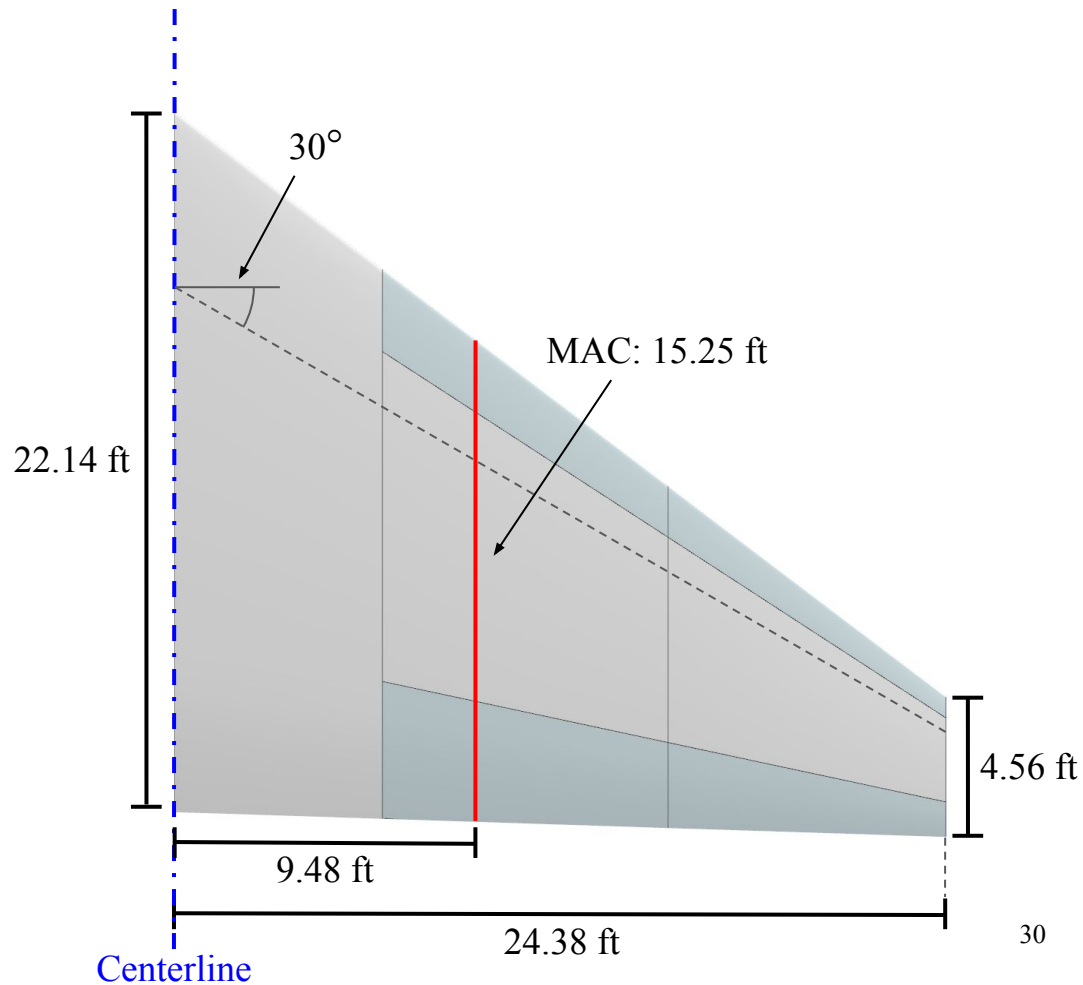
- Trapezoidal wing with taper, sweep, and twist

Dimensions

- Span: 48.75 ft
- Reference Area: 648 ft²
- MAC: 15.25 ft
- Aspect Ratio: 3.67
- Taper Ratio: 0.2
- Sweep: 30°
- Tip Twist: -2.7°
- Incidence: 0°

Cruise Performance

- $C_{D,ind} = 160$ counts, $C_{D,visc} = 137$ counts



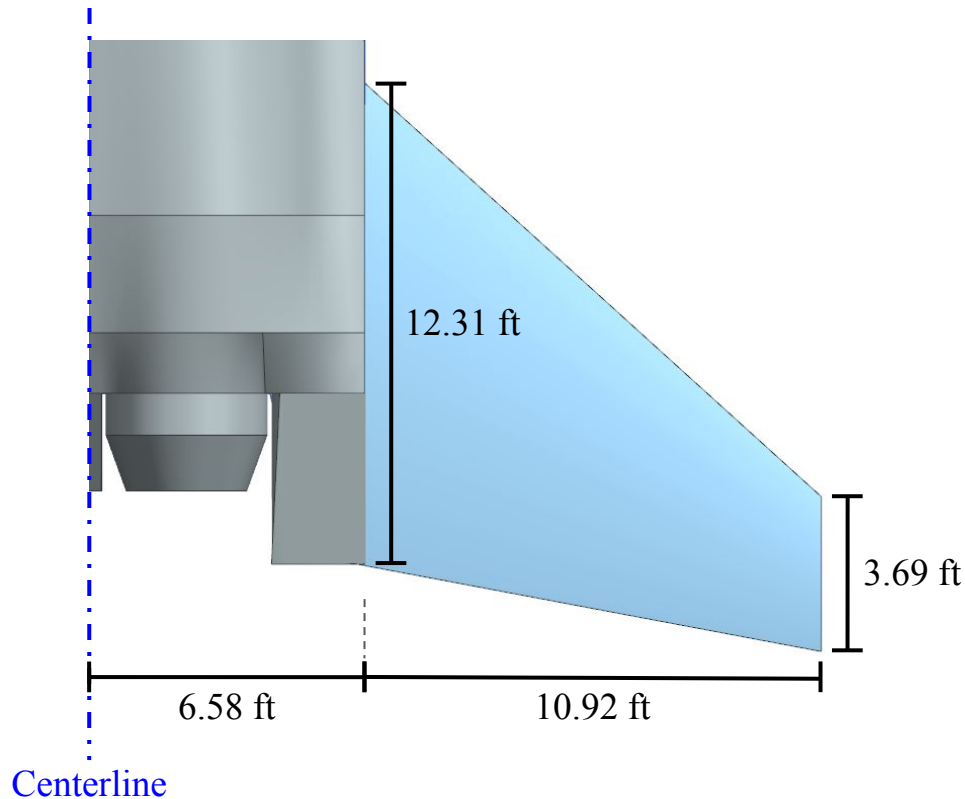
Empennage Design

Sizing

- Raymer tail volume coefficients [4]
 - $c_{HT} = 0.36$
 - $c_{VT} = 0.07$
- Swept to achieve 5° higher LE sweep than main wing

Dimensions

- Horizontal Tail
 - Moment Arm Length: 18.5 ft
 - Taper Ratio: 0.3
 - Aspect Ratio: 3
- Vertical Tail
 - Moment Arm Length: 13.2 ft
 - Taper Ratio: 0.4
 - Aspect Ratio: 1.4



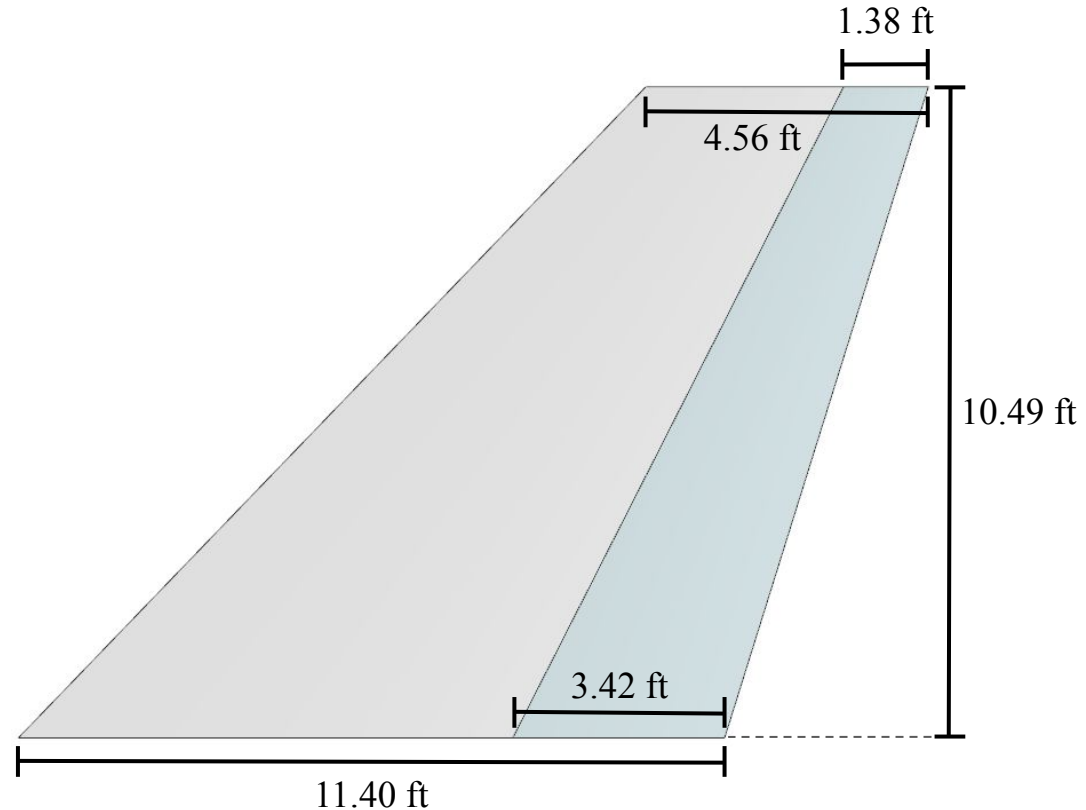
Control Surfaces: Elevator and Rudder

Elevator

- All-moving stabilator
 - Improves maneuverability
 - Reduces c_{HT} by 10%

Rudder

- 30% chord fraction
 - Inboard chord: 3.42 ft
 - Outboard chord: 1.38 ft



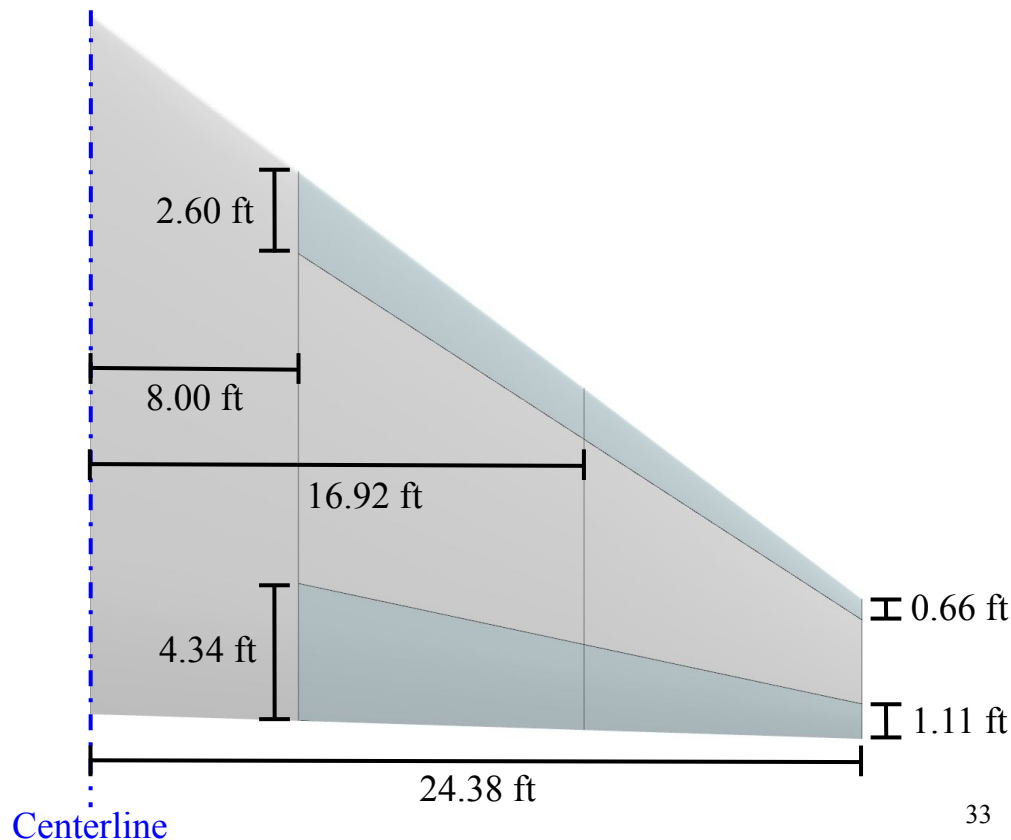
Control Surfaces: Flaps and Ailerons

Flaps

- Leading edge plain flaps
 - Span: 27% to 100%
 - Chord fraction: 15%
- Trailing edge slotted flaps
 - Span: 27% to 64%
 - Chord fraction: 25%
 - Increase takeoff $C_{L,max}$ for catapult launch

Slotted Flaperons

- Span: 64% to 100%
 - Begins at hinge point
- Chord fraction: 25%
 - Matches Raymer recommendations [4]



Airfoil Design

Multi-Point Design Optimization

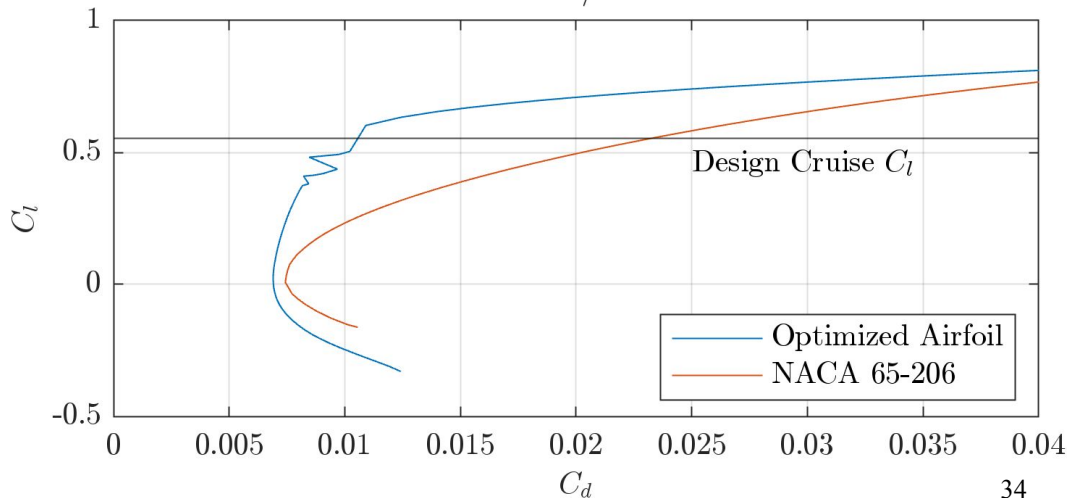
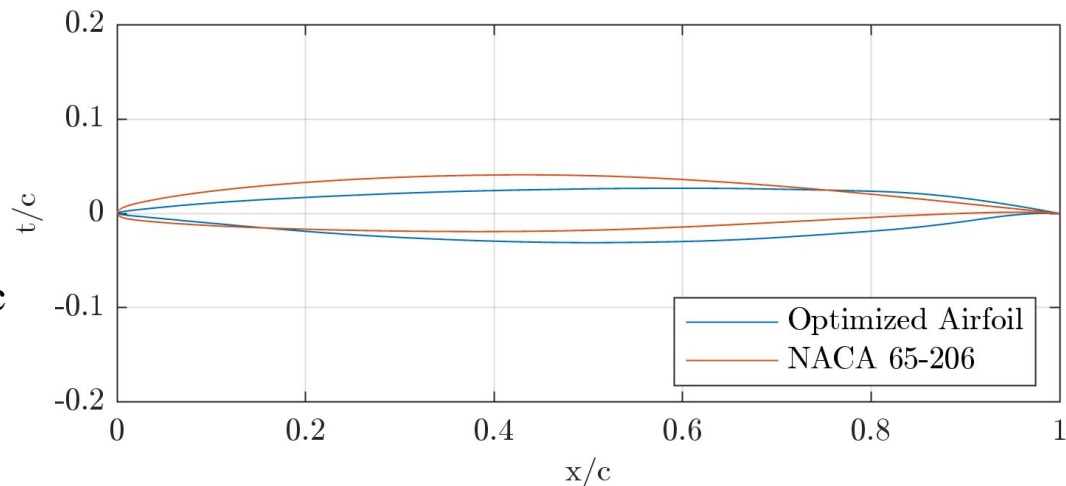
- Drag reduction at cruise and supersonic dash operating conditions

Results

- Flattened top surface and aft camber reduces cruise wave drag
- Thin leading edge reduces $C_{d, \text{wave}}$
- Sectional drag reduced by 127 counts

Tail Airfoils

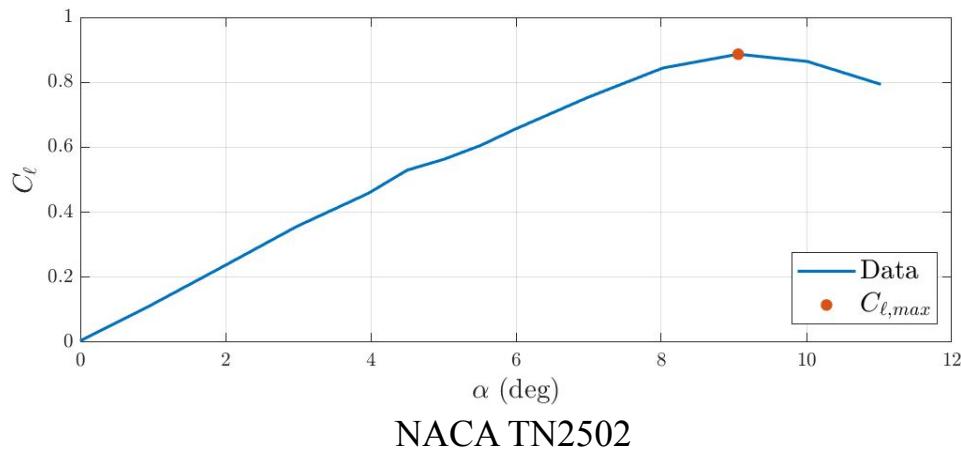
- Biconvex airfoil offers low $C_{d, \text{wave}}$



$C_{L,max}$ Estimation

Clean $C_{\ell,max}$

- XFOIL and ADflow results suspect due to sharp leading edge and unsteady flow
- NACA TN2502 experimental data used
 - $C_{\ell,max} = 0.89$ for NACA 64A006



Flapped $C_{\ell,max}$

- Raymer provides $\Delta C_{\ell,max}$ values
- Approach assumed half of takeoff

Aircraft $C_{L,max}$

- Critical section method used in AVL for each configuration

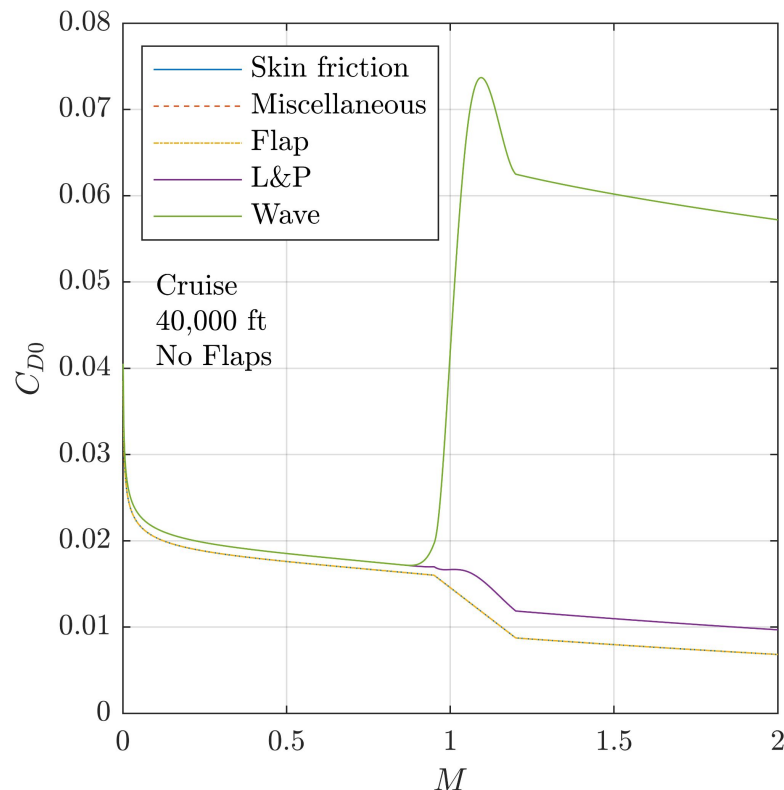
Configuration	$C_{L,max}$	Stall Speed (kts)
Clean (Cruise)	0.79	351
Approach	1.11	131
Takeoff	1.42	144

Parasitic Drag Model

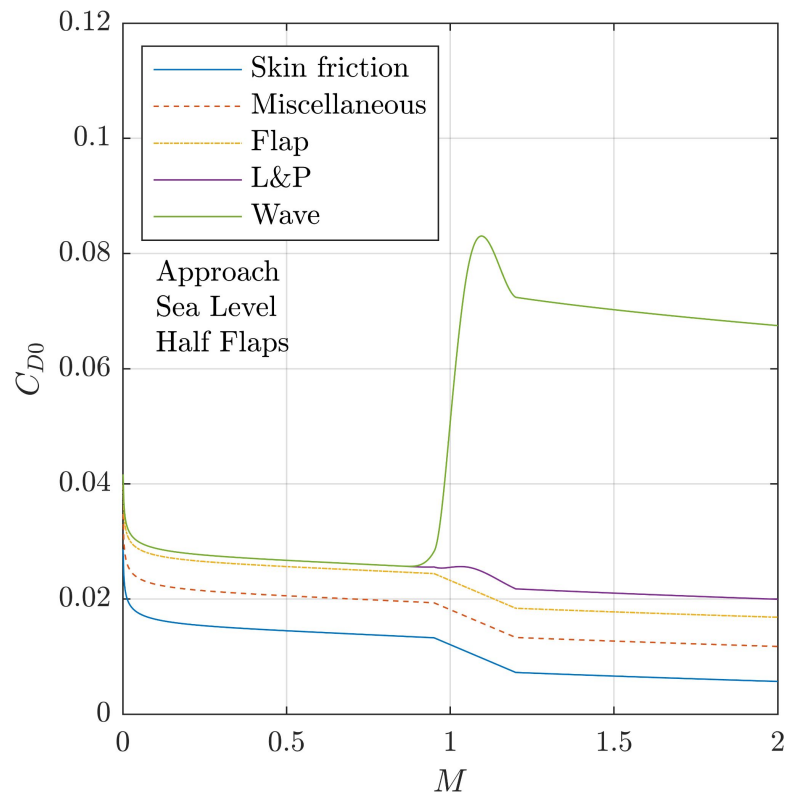
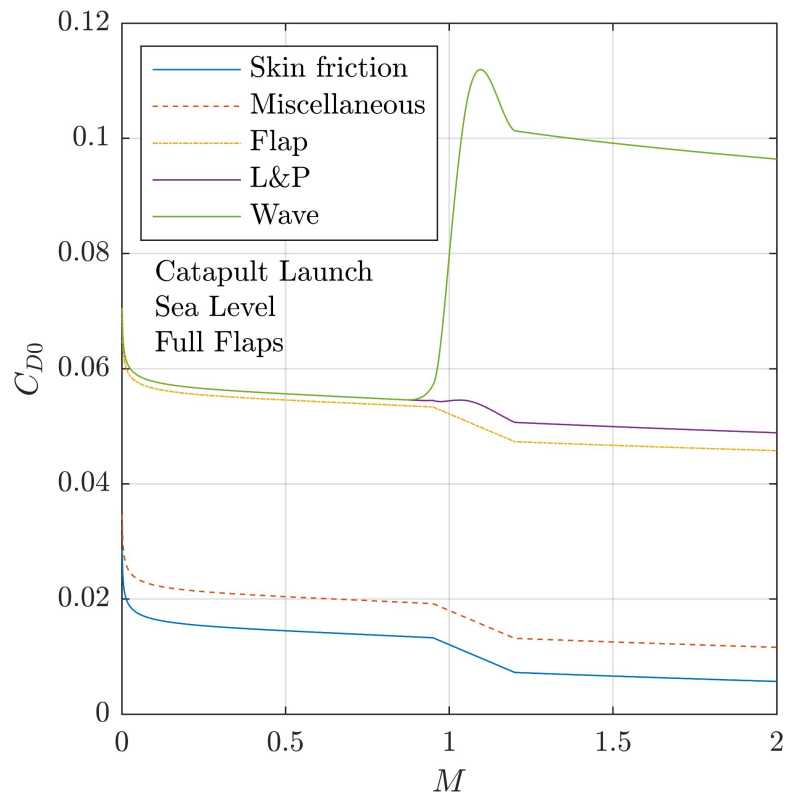
Component Drag Estimation

- Varies with Mach, Re, altitude
- Leakages and protuberance drag, 5% of total

Components	
Skin friction	Miscellaneous
Wing	Landing gear
Horz. Tail	Arresting hook
Vert. Tail	
Fuselage	
Canopy	



Parasitic Drag Model

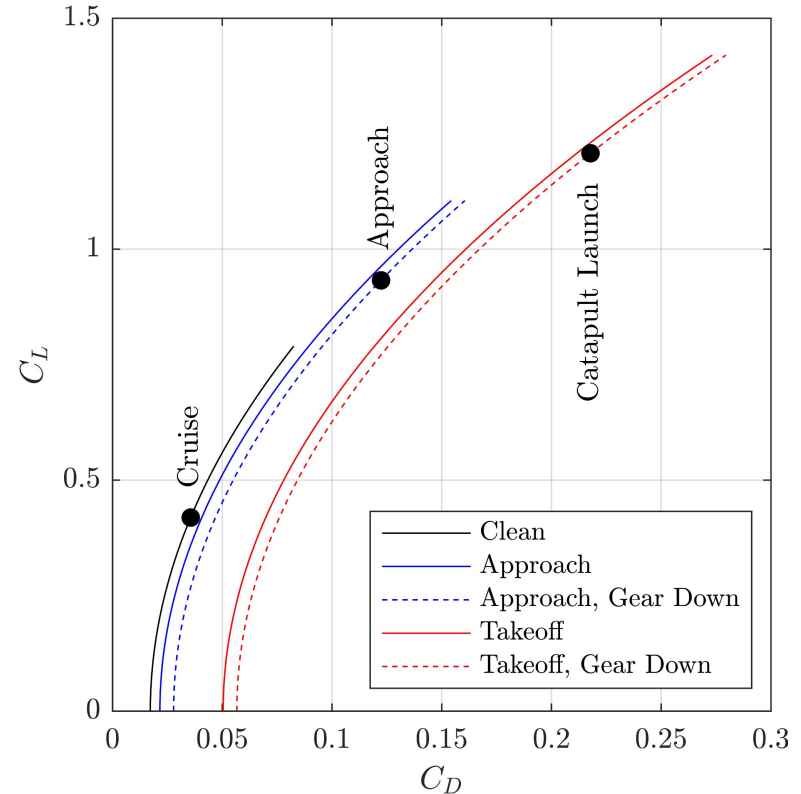


Drag Polars

Assumptions

- Trim drag assumes 5% increase in K

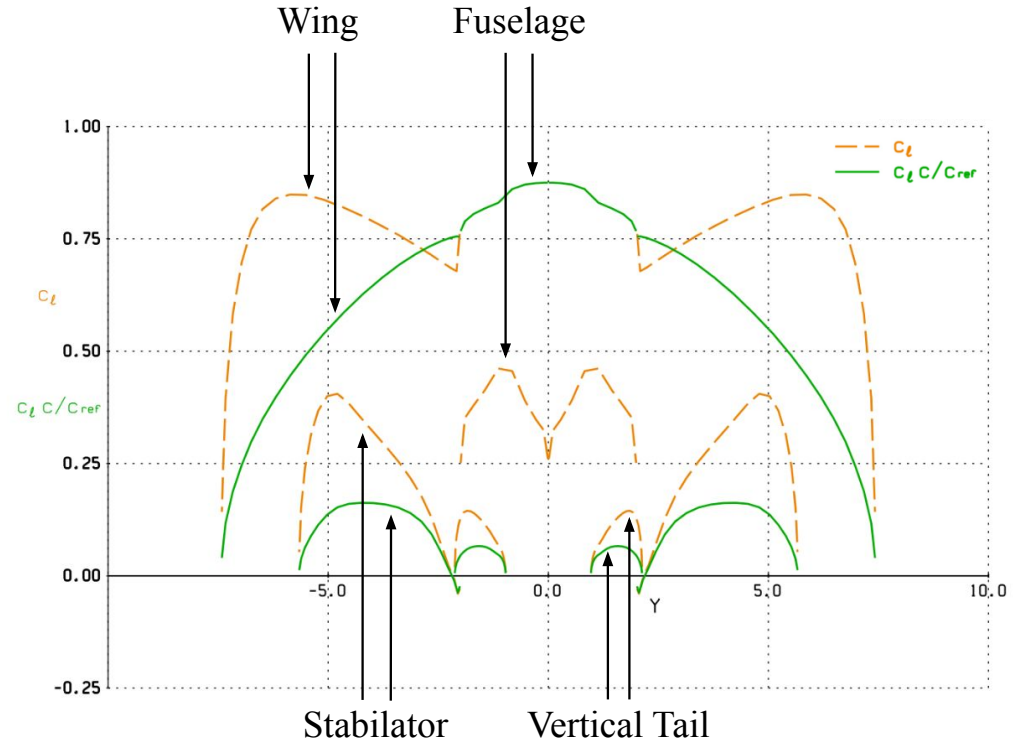
Polar	C_{D0}	e	$C_{L,max}$
Clean	0.0173	0.83	0.79
Approach	0.0216	0.80	1.11
Approach, gear down	0.0279	0.80	1.11
Takeoff	0.0504	0.78	1.42
Takeoff, gear down	0.0567	0.78	1.42



Aerodynamic Loads: Clean

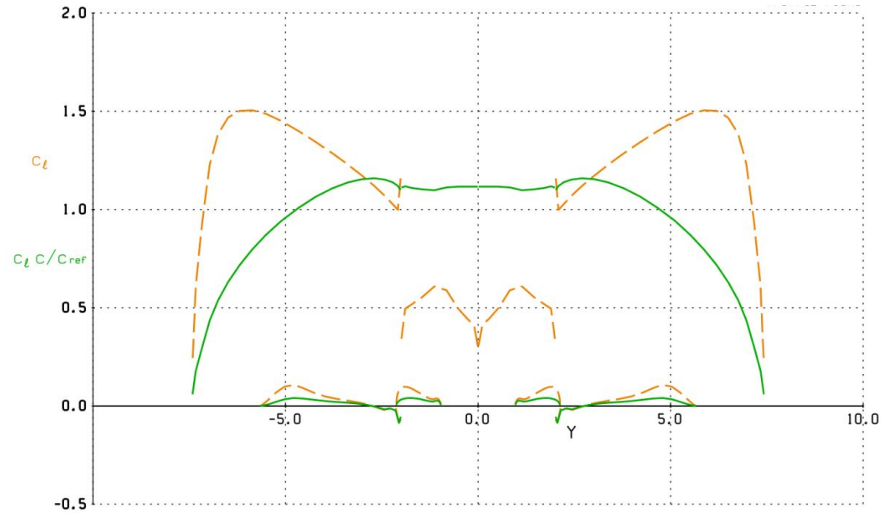
Load Distribution

- Found using AVL at $C_{L,max}$
- Force proportional to $C_\ell \cdot c/c_{ref}$
- C_ℓ distribution shows stall location

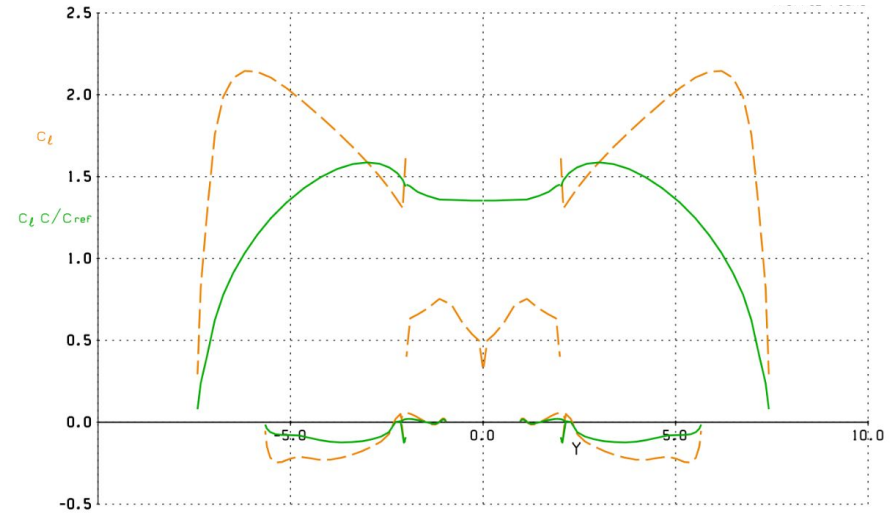


Clean, $C_L = 0.79$

Aerodynamic Loads: Approach and Takeoff



Approach, $C_L = 1.11$



Takeoff, $C_L = 1.42$

Flaps move the lift distribution outboard.

Empty Mass Budget

Installed Engine components

- Engine
- Engine Mounts and Section
- Cooling Systems
- Starter (pneumatic)

All-else empty components

- Avionics, Flight and Engine Controls
- Instruments
- Hydraulics and Electrical
- Furnishings
- Air Conditioning
- Handling Gear
- Fuel Systems
- Air Induction

Component	Weight (lb)
Fuselage	5,018.2
Wing	4,774.9
Stabilator	510.5
Vertical Tail	1,173.7
Nose Gear	626.1
Main Gear	1,044.8
Installed Engine	8,941.5
All-Else Empty	9,117.8
Total	31,207

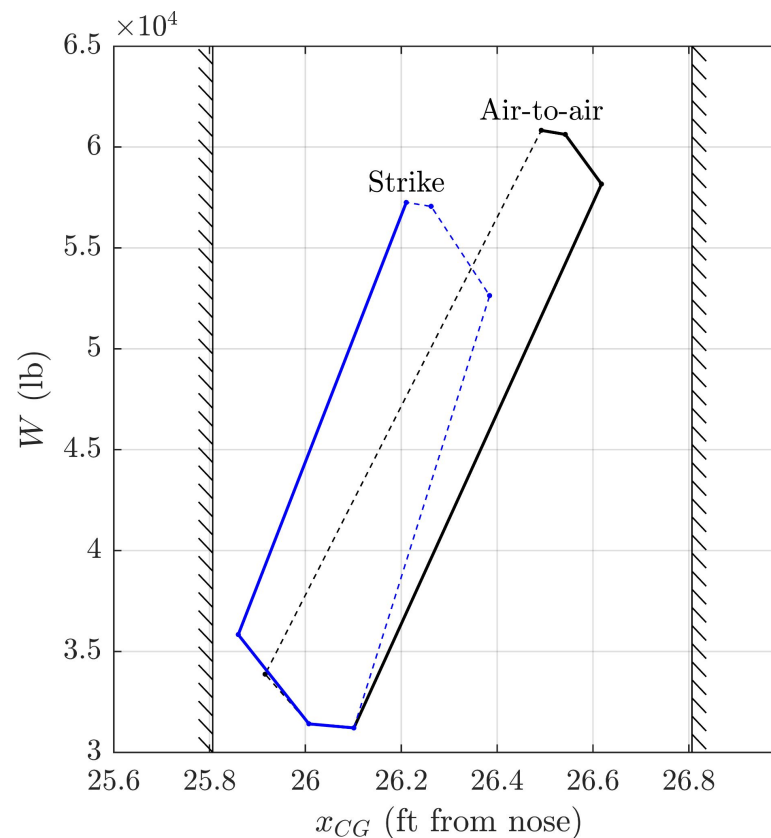
CG Loading Diagrams

CG Limits

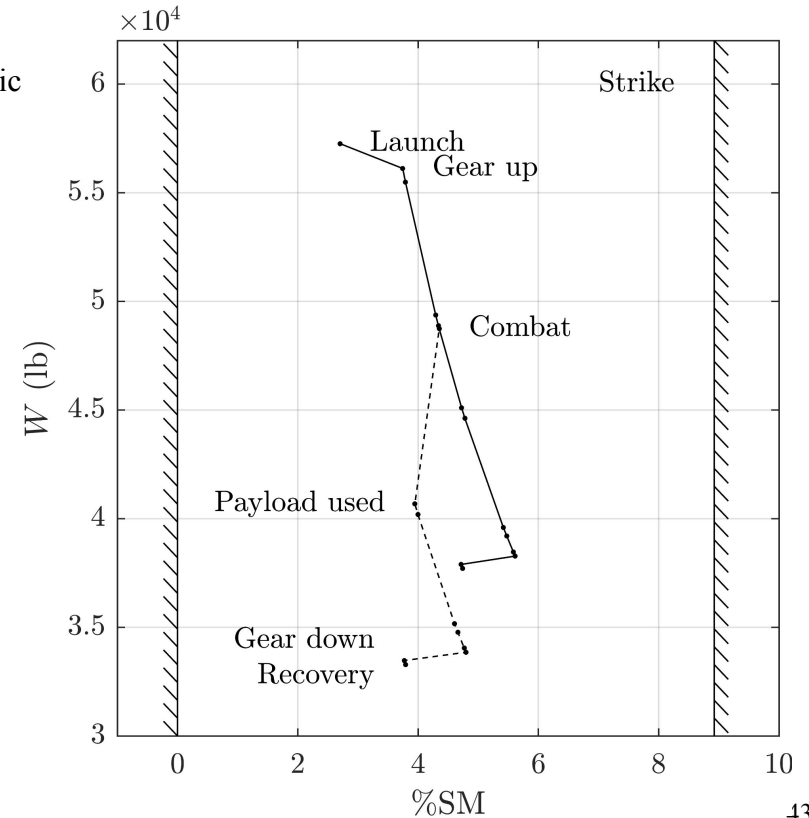
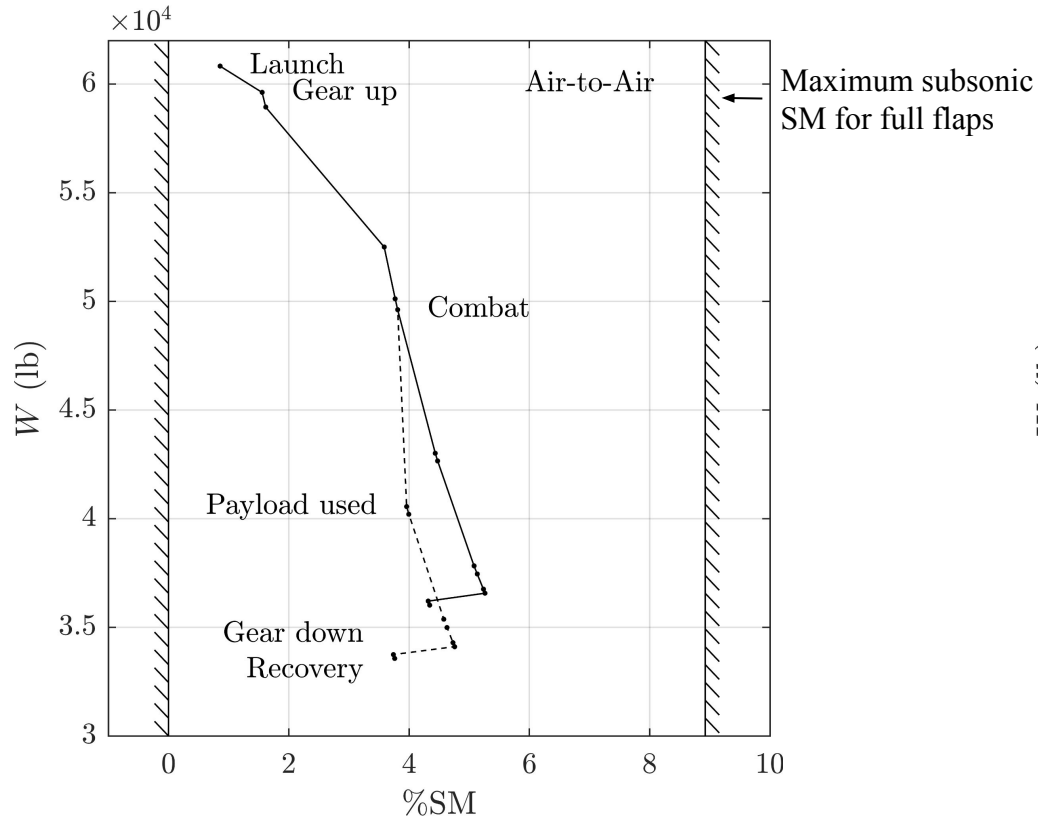
- Forward limit
 - < 20% load on nose wheel
- Aft limit
 - > 25° longitudinal tip-over angle

Loading Order Permutations

- Empty weight
- Crew
- Fuel
- Payload

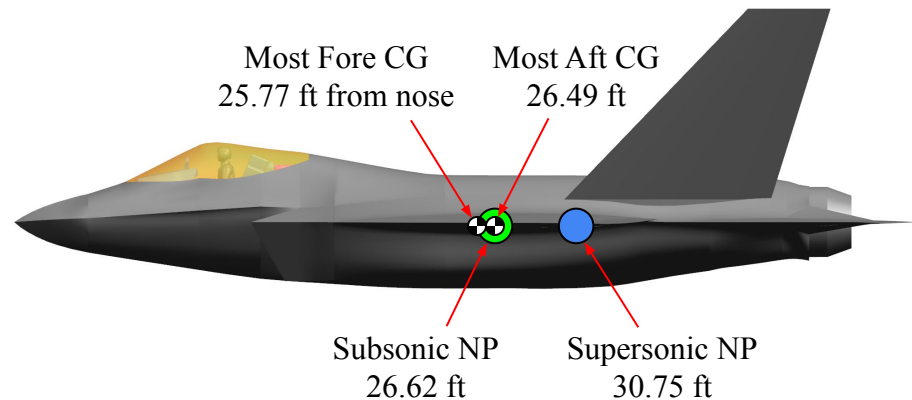


CG and Static Margin Envelope



Longitudinal Static Stability

Subsonic	Air to Air Static Margin (%MAC)	Strike Static Margin (%MAC)
Most Fore	0.86	2.7
Most Aft	5.3	5.6
Supersonic	Air to Air Static Margin (%MAC)	Strike Static Margin (%MAC)
Most Fore	28	30
Most Aft	32	33



Maximum cruise SM: 34%

High supersonic SM requires fuel management system.

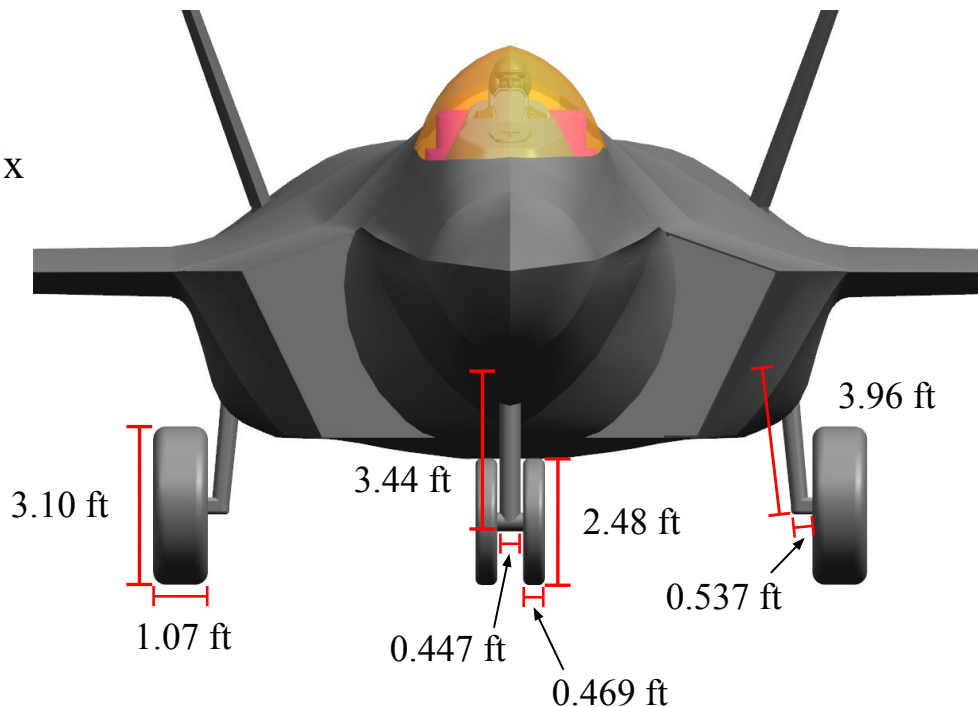
Landing Gear Design

Strut Sizing

- Main gear stroke height: 1.25 ft
- Nose gear stroke height: 1.30 ft
- Minimum strut lengths based on 2.5x stroke length from Raymer [4]
- Minimum main strut length: 3.12 ft
- Minimum nose strut length: 3.26 ft

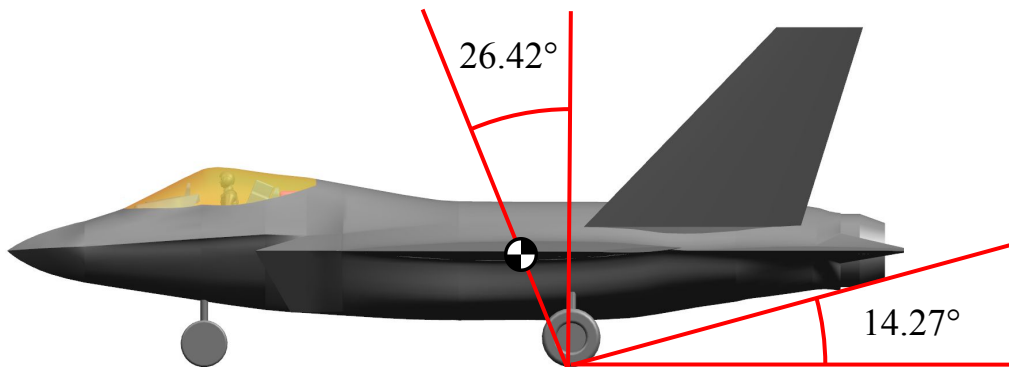
Wheel Sizing

- Based on maximum static and dynamic loads from Raymer [4]

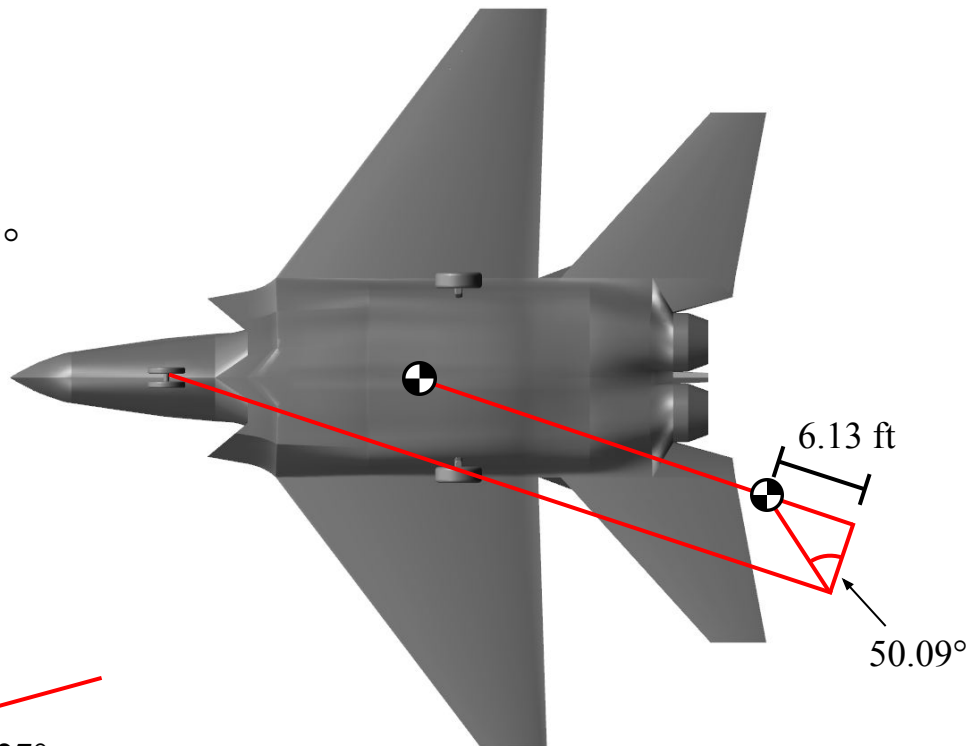


Landing Gear Angles

- Tipback: **14.27°**
- Longitudinal tip-over (aft C.G): **26.42°** > 25°
- Overturn: **50.09°** < 54°



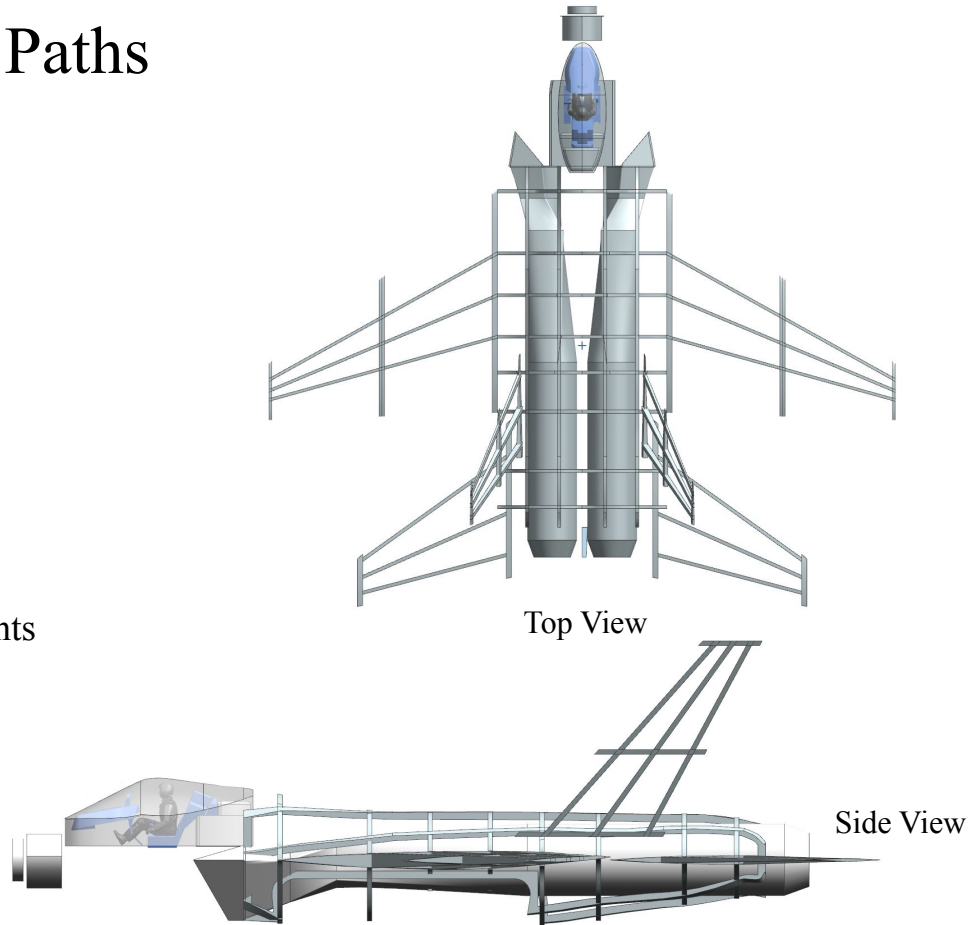
Side View



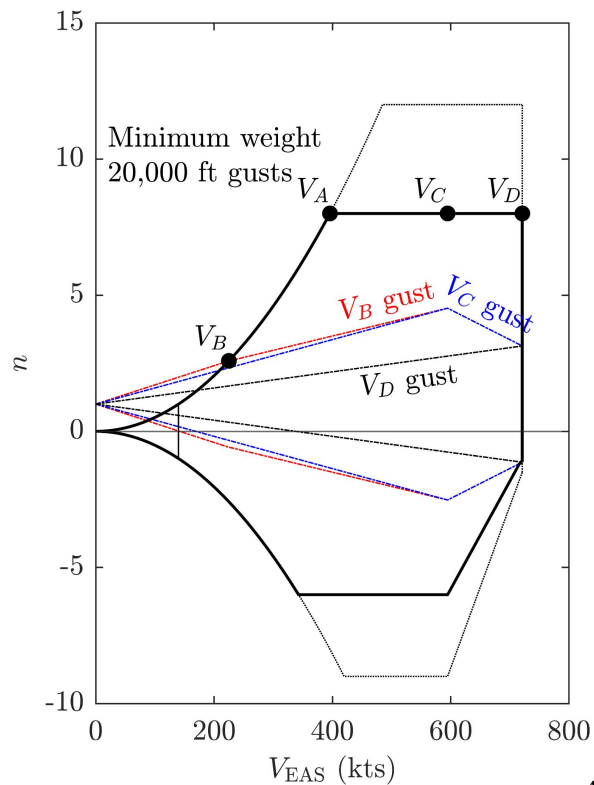
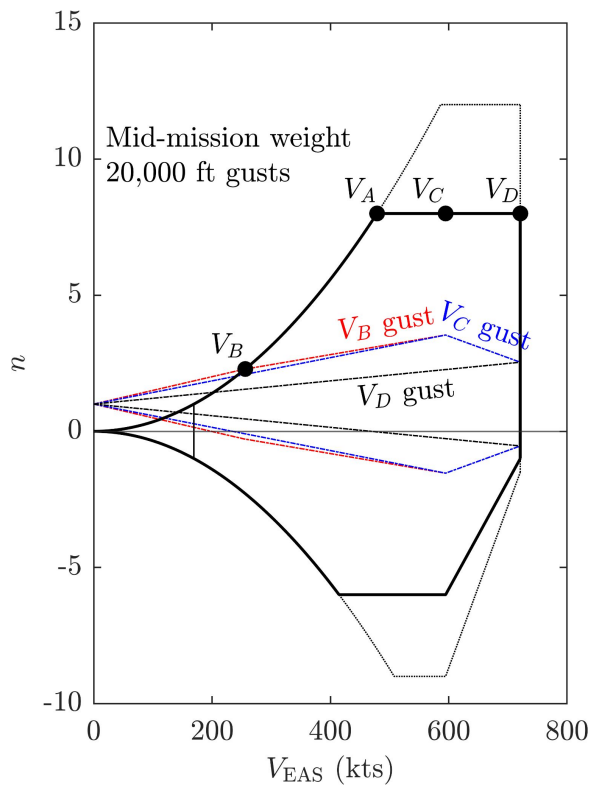
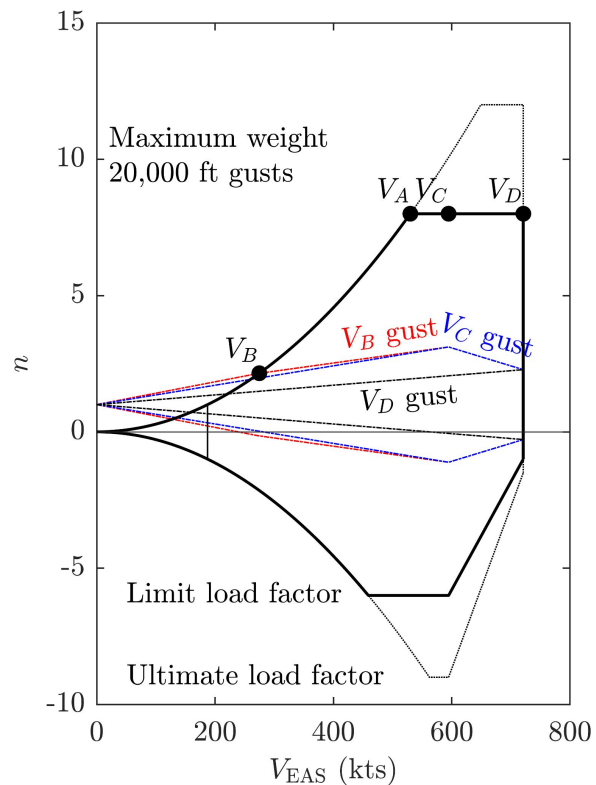
Bottom View

Internal Structure and Load Paths

- Three-spar configuration in wingbox, stabilators, and vertical tail
 - Torsional support
 - Ribs at end-points
- Fuselage consists of 8 ring-frames
 - Buckling and deformation support
- Fuselage Longerons
 - Bending and deformation support
 - Main fuselage load bearing components



V-n Diagrams



Air-to-Air Combat Performance

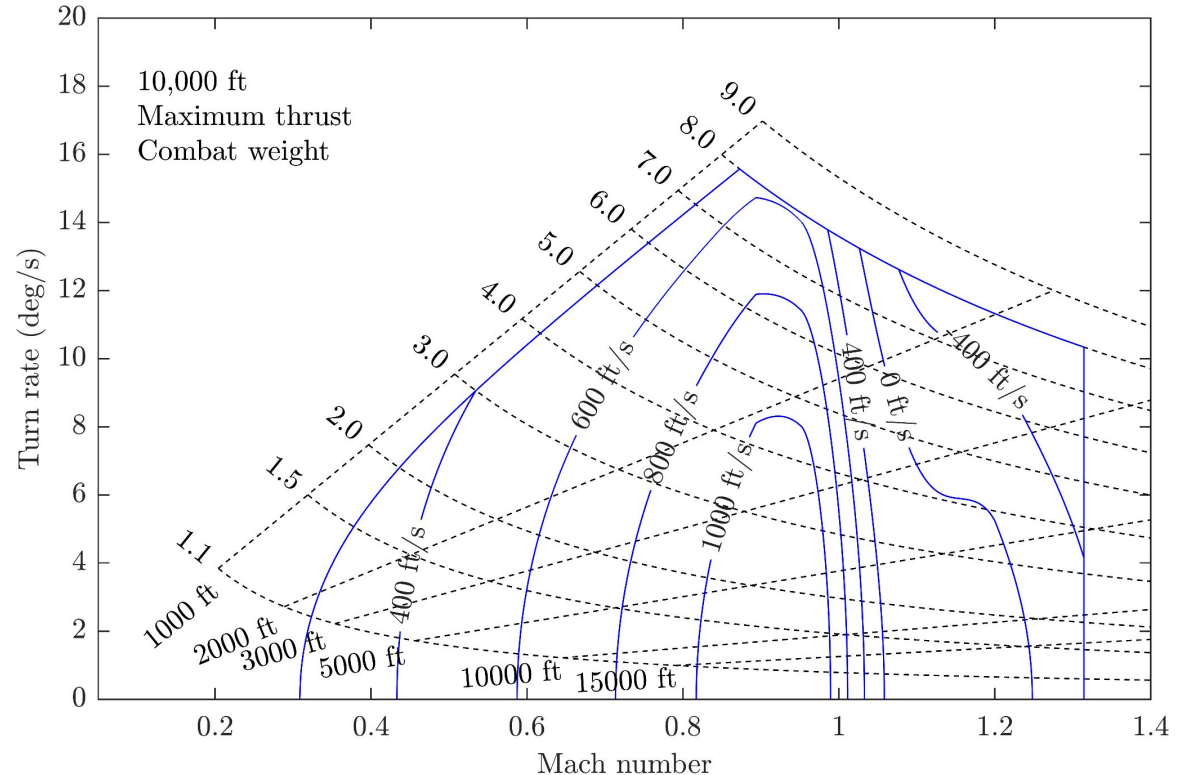
Combat Maneuverability

- Sustained turn rate: 13 deg/s
- Maximum turn rate: 15 deg/s
- Limit load factor: 8g

Sustained Turn Requirement

- 20,000 ft
- Sustained turn rate: 12 deg/s

Satisfies RFP requirements



Launch and Recovery Performance

Catapult Launch

- Takeoff SEROC prevents 10 ft sink
- $C_L = 1.21 < 0.9C_{L,max} = 1.28$
- $> 0.065g$ longitudinal acceleration
- Rudder provides sufficient OEI control

Recovery

- Approach speed < 145 kts, 15 kts WOD
- Landing weight: 39,177 lb
 - Includes 25% fuel weight, 50% store weight
- Hookload: 145,000 lb

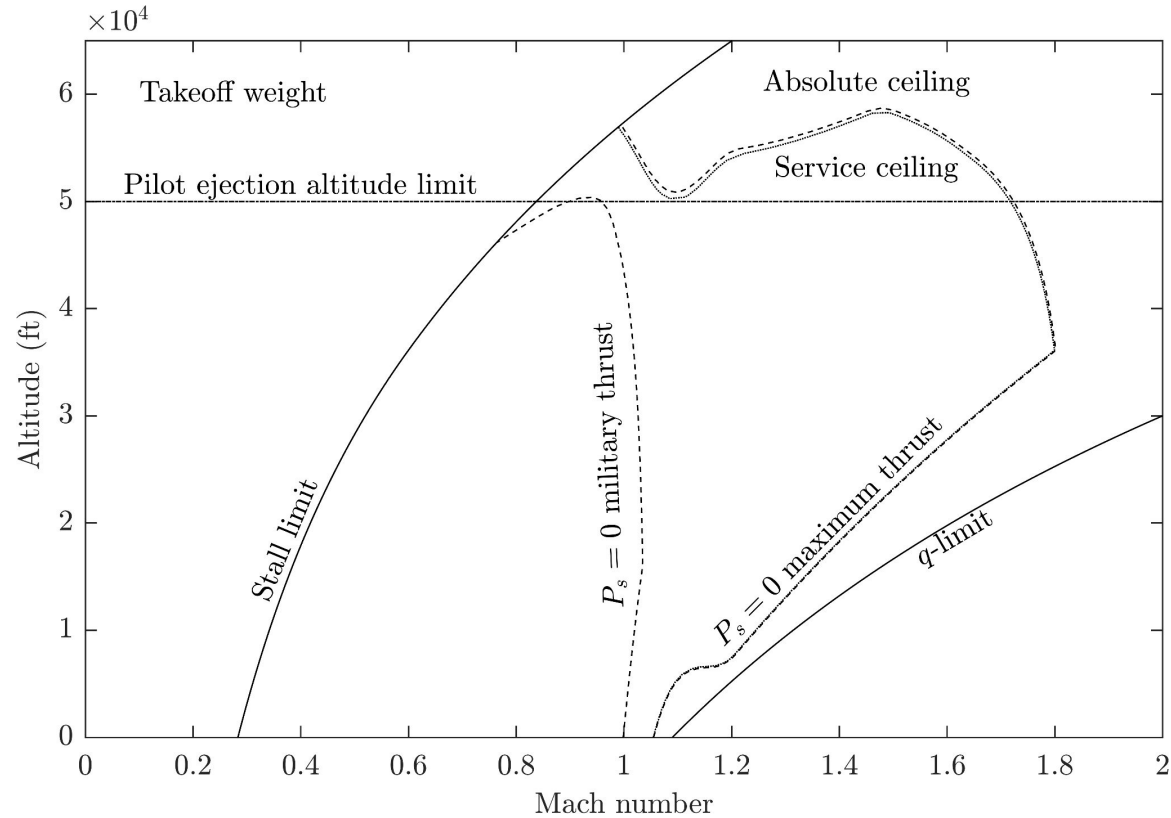
Catapult End Speed (kts)	Peak Acceleration (g's)	Stall Speed (kts)	Approach Speed	Engaging Speed
156	4	116	127	134

SEROC

- Takeoff: > 200 ft/min
- Approach: > 500 ft/min

Takeoff SEROC (ft/min)	Approach SEROC
3198	5253

Flight Envelope



Conclusion

Performance and Objective

- Lower unit cost than F/A-18E/F: 98.8 \$M 2025
- Exceeds minimum performance requirements
- Satisfies RFP requirements



References

- [1] *Standard Aircraft Characteristics F/A-18E Super Hornet*. NAVAIR00-110AF18-6. Boeing. 2001
- [2] Joaquim R.R.A. Martins. *The Metabook of Aircraft Design*. Dr. Martins, Joaquim R.R.A., 2021.
- [3] Leland M. Nicolai. *Fundamentals of Aircraft Design*. American Institute of Aeronautics and Astronautics, 2010. isbn: 978-1-60086-751-4.
- [4] Daniel P. Raymer. *Aircraft Design: A Conceptual Approach*. American Institute of Aeronautics and Astronautics, 2018. isbn: 9781624104909
- [5] Jan Dr. Roskam. *Airplane Design*. Roskam Aviation and Engineering Corporation, 1985.
- [6] Airport Data. 2023. url: https://www.faa.gov/air_traffic/flight_info/aeronav/aero_data/Airport_Data/.
- [7] F/A-18 corrosion maintenance doesn't consistently meet Navy and Marine Corps standards. 2021. url: <https://www.navytimes.com/news/your-navy/2021/10/05/fa-18-corrosion-maintenance-doesnt-consistently-meet-navy-and-marine-corps-standards/#:~:text=Your%20Navy-,F/A%2D18%20corrosion%20maintenance%20doesn%27t%20consistently%20meet,of%20Defense%20Inspector%20General%20report.>
- [8] MCDONNELL DOUGLAS F/A-18 HORNET. 2018. url: <https://janes.migavia.com/usa/mdc/f-18.html>.
- [9] List of military electronics of the United States. 2025. url: https://en.wikipedia.org/wiki/List_of_military_electronics_of_the_United_States.
- [10] AN/APG-81. 2016. url: <https://www.deagel.com/Components/ANAPG-81/a001381>.
- [11] AN/ALQ-214. 2017. url: <https://www.australiandefence.com.au/c4i-ew/harris-to-supply-super-hornet-ew-countermeasures>.
- [12] From the JED Archives: Protect and Strike – NAVAIR/PEO-T Advances Naval Airborne EW. 2023. url: https://www.jedonline.com/2023/12/12/from-the-jed-archives-protect-and-strike-navair-peo-t-advances-naval-airborne-ew/?print=print&utm_source.
- [13] \$400,000 helmet turns fighter jets transparent for 360-degree vision. 2024. url: <https://newatlas.com/aircraft/f-35-lightning-gen-3-hdms-helmet-cost/#:~:text=It%20took%20a%20decade%20of,4%2Dyear%20degree%20from%20Harvard>.
- [14] Onboard Oxygen Generation System Market. 2025. url: <https://pmarketresearch.com/auto/onboard-oxygen-generation-system-market/>.
- [15] F-35 Gen III Helmet Mounted Display System (HMDS). 2022. url: <https://www.collinsaerospace.com/what-we-do/industries/military-and-defense/displays-and-controls/airborne/helmet-mounted-displays/f-35-gen-iii-helmet-mounted-display-system>

Backup Slides

List of Symbols

$C_{L,max}$	Maximum lift coefficient
T/W	Thrust-To-Weight Ratio
AR	Aspect Ratio
W/S	Wing Loading (lbs/ft ²)
S	Wing reference area (ft ²)
T	Thrust (lbs)
$C_{D,ind}$	Induced drag coefficient
$C_{D,visc}$	Viscous drag coefficient
C_{HT}	Horizontal tail volume coefficient
C_{VT}	Vertical tail volume coefficient
$C_{D,wave}$	Wave drag coefficient
$CD0$	Zero-lift drag coefficient
e	Oswald efficiency
K	$1/\pi ARe$
C_D	Drag coefficient
C_L	Lift coefficient
C_l	Sectional lift coefficient
c	Chord
c_{ref}	Reference chord
W	Weight
x_{CG}	Position from the center of gravity

Full Aircraft Comparison Tables

Aircraft	Takeoff Weight (lb)	Empty Weight (lb)	Air to air mission payload (lb)	Strike mission payload (lb)	Combat Radius (nmi)
F/A-67	31207	60826	2460	4424	1000
F/A-18E	66000	32081	2946	5442	444-462
F-35C	70000	34800	up to 18000	up to 18000	670
F-15E	81000	37500	2460	4398	687

Aircraft	Air to air mission fuel burn (lb)	Strike mission fuel burn (lb)	Cruise/Landing/ Takeoff CL	Cruise L/D	T/W	W/S (lb/ft^2)
F/A-67	26959	21425	0.79/1.11/1.42	11.7	0.98	94
F/A-18E	18218	18313	-	-	0.65	132
F-35C	18600	18600	-	-	0.61	105
F-15E	22300	22300	-	-	0.73	133

Full Aircraft Comparison Tables

Aircraft	Engine Type	Maximum sea-level static thrust (per engine)	Engine Cruise SFC (1/s)
F/A-67	F110-GE-132	33093	0.68
F/A-18E	F414-GE-400	21496	0.82
F-35C	F135-PW-400	43000	-
F-15E	F110-GE-129	29474	0.67

Aircraft	Span (ft)	Reference Area (ft^2)	Aspect Ratio	Average wing t/c	Cruise Mach number (maximum)	Cruise Mach number (economic)
F/A-67	48.8	647.5	3.67	5.7%	1.8	0.84
F/A-18E	44.7	500	3.5	-	1.6	0.84
F-35C	43	668	2.7	-	1.6	-
F-15E	42.8	608	3.0	-	2.5	-

Aircraft Comparison Tables

Aircraft	Static margin (Subsonic/Supersonic) (Most Fore, Air-to-Air)	Maximum landing distance (ft)	Maximum takeoff distance (ft)	Flyaway cost for unit production run of 500 (USD 2025 \$M)
F/A-67	0.86%/28%	6,800	4,800	98.8
F/A-18E	-	8,500	4,666	113
F-35C	-	-	-	127
F-15E	-	12,000	11,000	61.4

Notes

- For landing, assume takeoff weight, altitude of 4,000 ft, temperature of 89F

Engine Comparison Table [3]

Engine	Max Thrust (SLS-lb)	Intermediate Thrust (SLS-lb)	TSFC Max (1/s)	TSFC Intermediate (1/s)
F110-GE-132	33,093	17,669	1.90	0.68
F414-GE-400	21,496	14,447	1.84	0.82
F110-GE-129	29,474	17,595	1.85	0.67
F110-PW-229	29,100	17,800	2.05	0.74
F110-PW-229A	32,500	20,100	1.86	0.71

Installed Thrust from Pressure Loss and Bleed

$$\%_{TL_{PR}} = c_r [\eta_{\text{ref}} - \eta_{\text{actual}}] * 100\%$$

$$\%_{TL_{EB}} = C_b \frac{\dot{m}_b}{\dot{m}_e}$$

$$T_{\text{installed}} = T \left(1 - \frac{\%_{TL_{PR}} + \%_{TL_{EB}}}{100} \right)$$

Thrust Lapse

$$\theta = T/T_{\text{std}} \quad \text{and} \quad \theta_0 = \theta \left(1 + \frac{\gamma - 1}{2} M^2 \right)$$

$$\delta = P/P_{\text{std}} \quad \text{and} \quad \delta_0 = \delta \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{\frac{\gamma}{\gamma - 1}}$$

$$\text{Maximum: } \begin{cases} \alpha = \delta_0, & \theta_0 \leq TR \\ \alpha = \delta_0(1 - 3.5(\theta_0 - TR)/\theta_0), & \theta_0 > TR \end{cases}$$

$$\text{Military: } \begin{cases} \alpha = \delta_0, & \theta_0 \leq TR \\ \alpha = \delta_0(1 - 3.8(\theta_0 - TR)/\theta_0), & \theta_0 > TR \end{cases}$$

Initial Weight Estimate from Historical Regressions

	Air to air	Strike
Empty weight (lb)	36,570	23,830
Payload weight (lb)	2,460	2,398
Fuel weight (lb)	40,800	24,420
Takeoff weight (lb)	80,040	50,840

Initial Weight Estimate from Initial Weight Code

	Air to air	Strike
Empty weight (lb)	43,379	43,341
Payload weight (lb)	2,460	4,424
Fuel weight (lb)	34,465	32,458
Takeoff weight (lb)	80,504	80,423

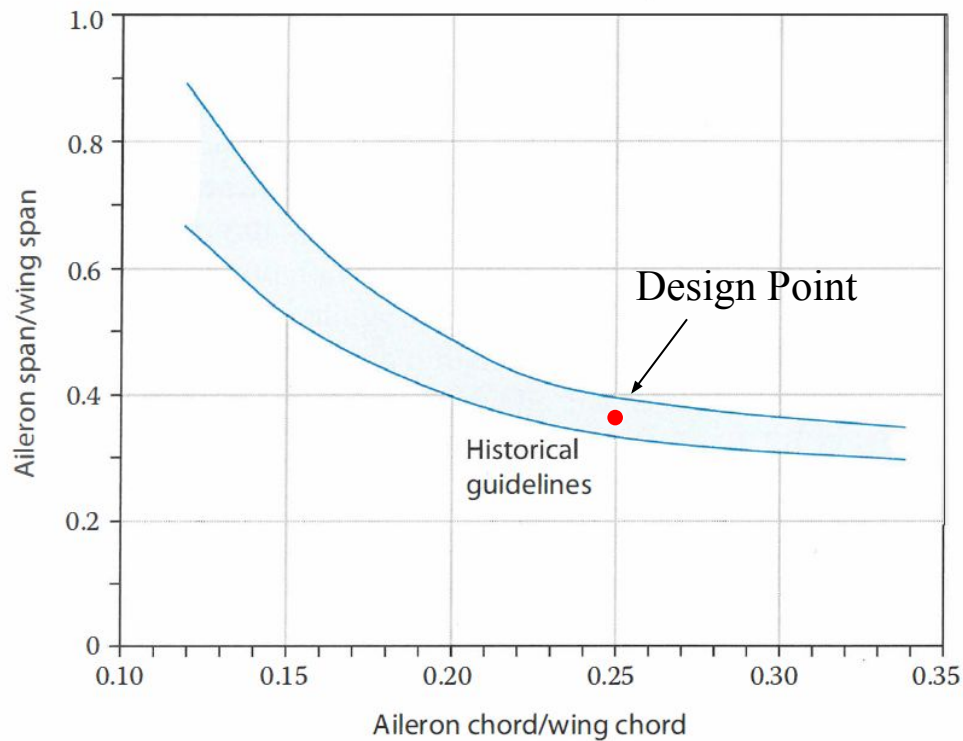
Weight Estimate from T-S Plot Design Point

	Air to air	Strike
Empty weight (lb)	43,117	43,254
Payload weight (lb)	2,460	4,424
Fuel weight (lb)	34,169	32,360
Takeoff weight (lb)	79,946	80,237

Gust Speeds at 20,000 ft

Gust	Speed (ft/s)
V_B (rough air)	66
V_C (at max design speed)	50
V_D (at max dive speed)	25

Aileron Sizing

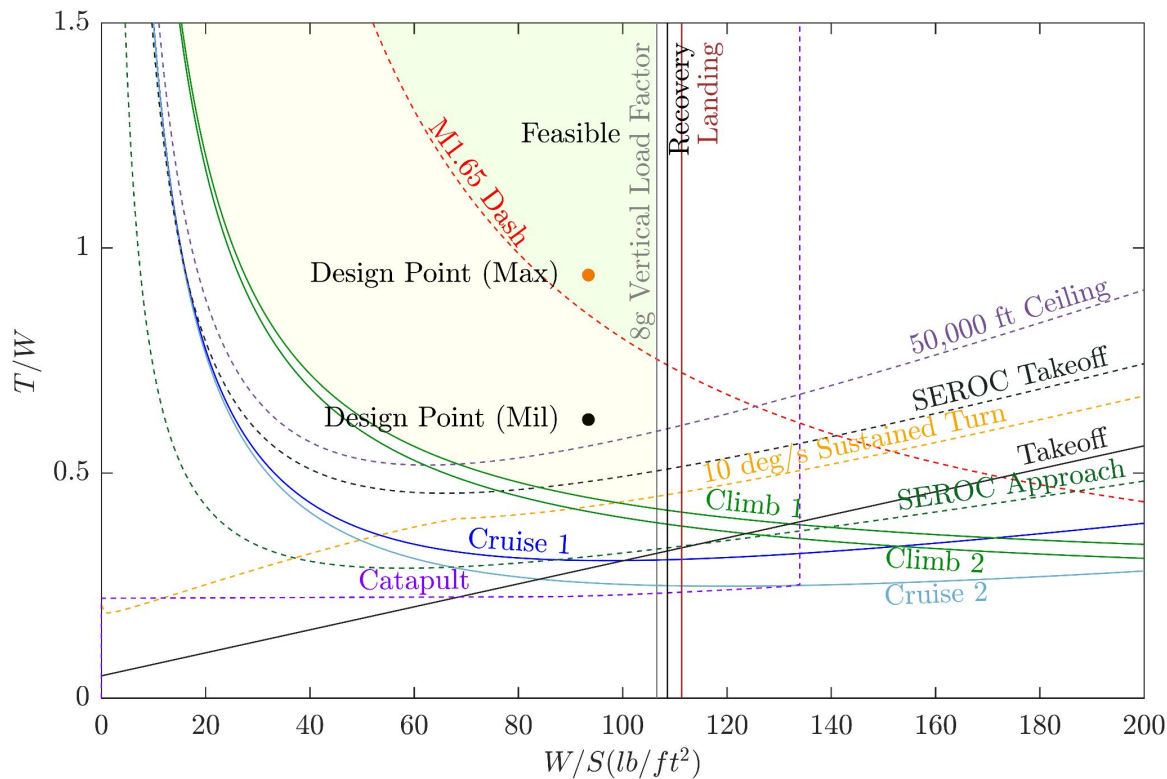


Raymer Fig. 6.3

Method Validation (Air-to-Air)

F/A-18E design point is
feasible for air to air mission.

	Weight (lb)	
	Predicted	Actual
Empty	28,891	30,554
Fuel	16,067	14,850
Takeoff	46,702	48,247

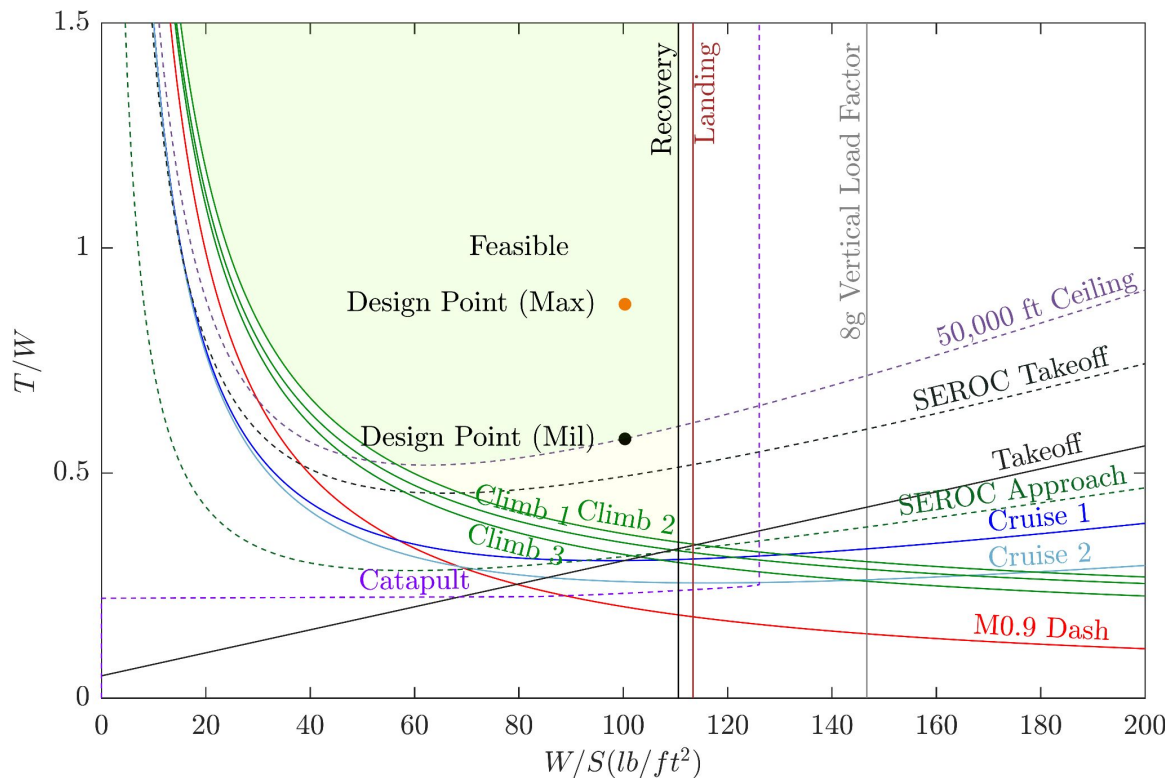


Method Validation (Strike)

F/A-18E design point is feasible for strike mission.

Large weight difference attributed to aerodynamic model inputs

	Weight (lb)	
	Predicted	Actual
Empty	28,891	30,554
Fuel	15,645	21,378
Takeoff	50,158	60,769



Energy Maneuverability at 20,000 ft

